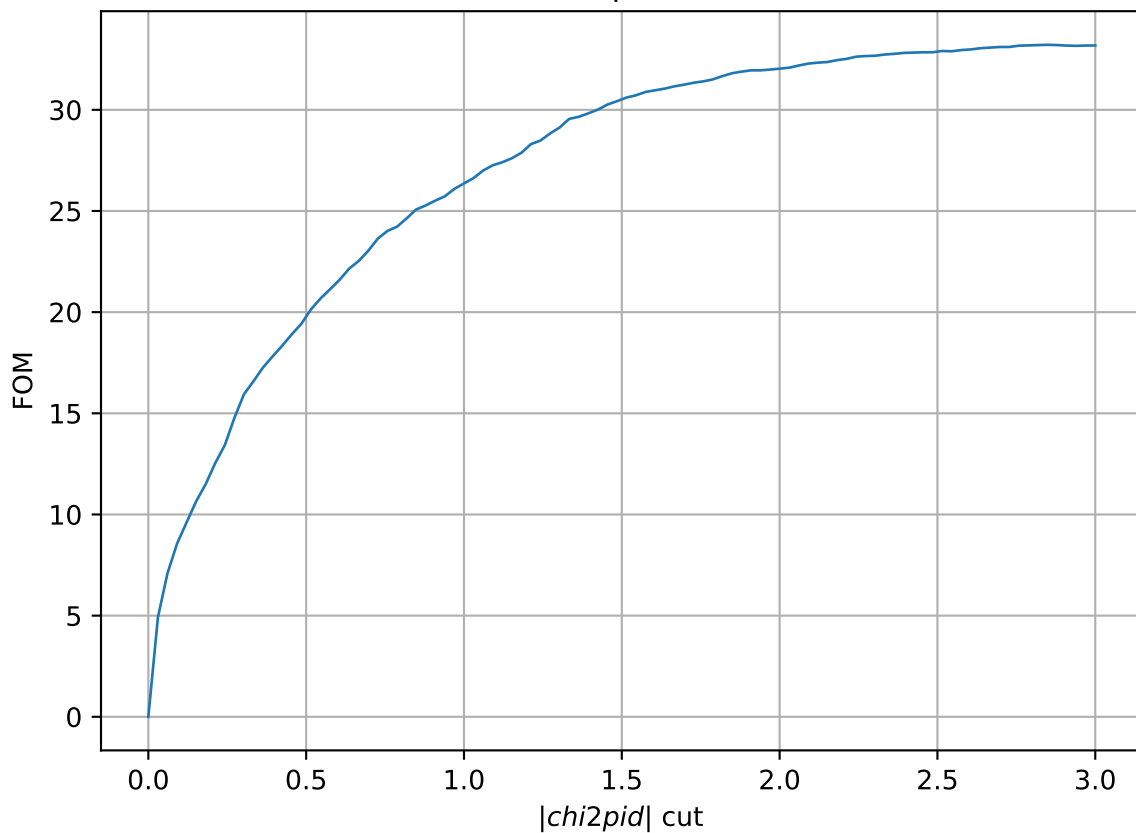
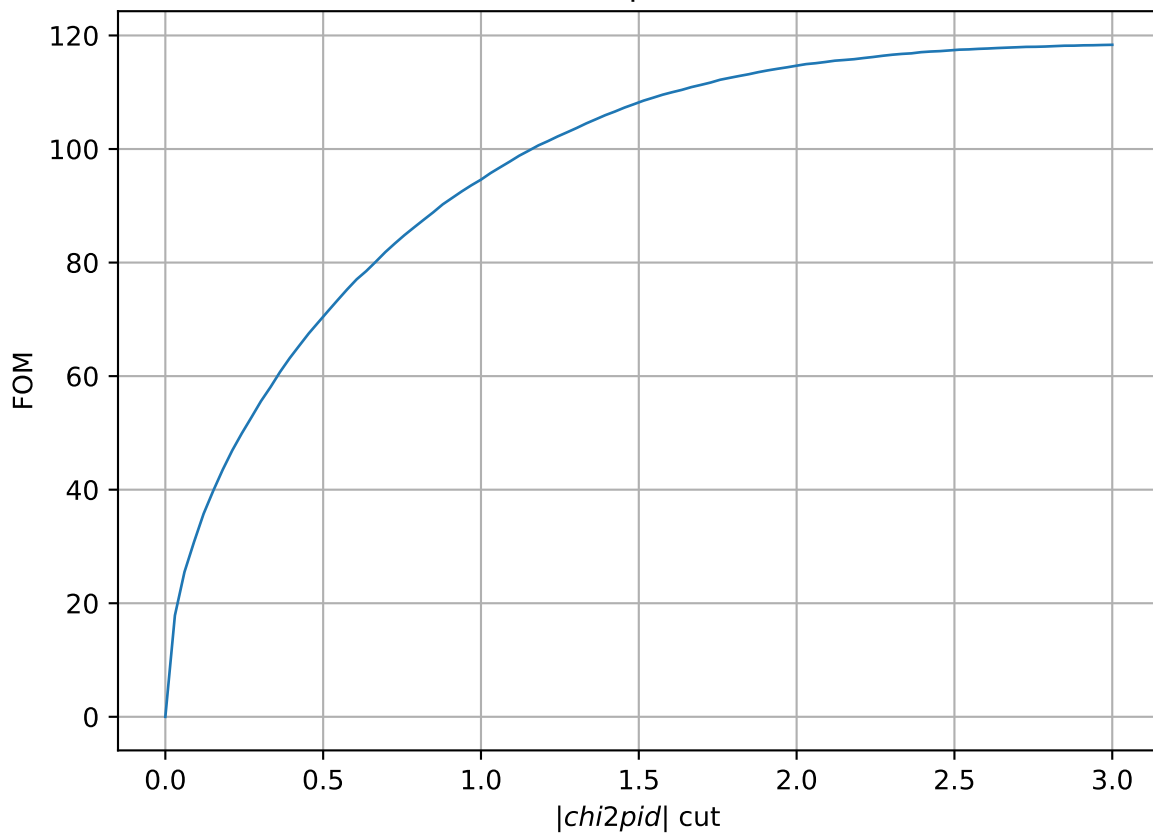


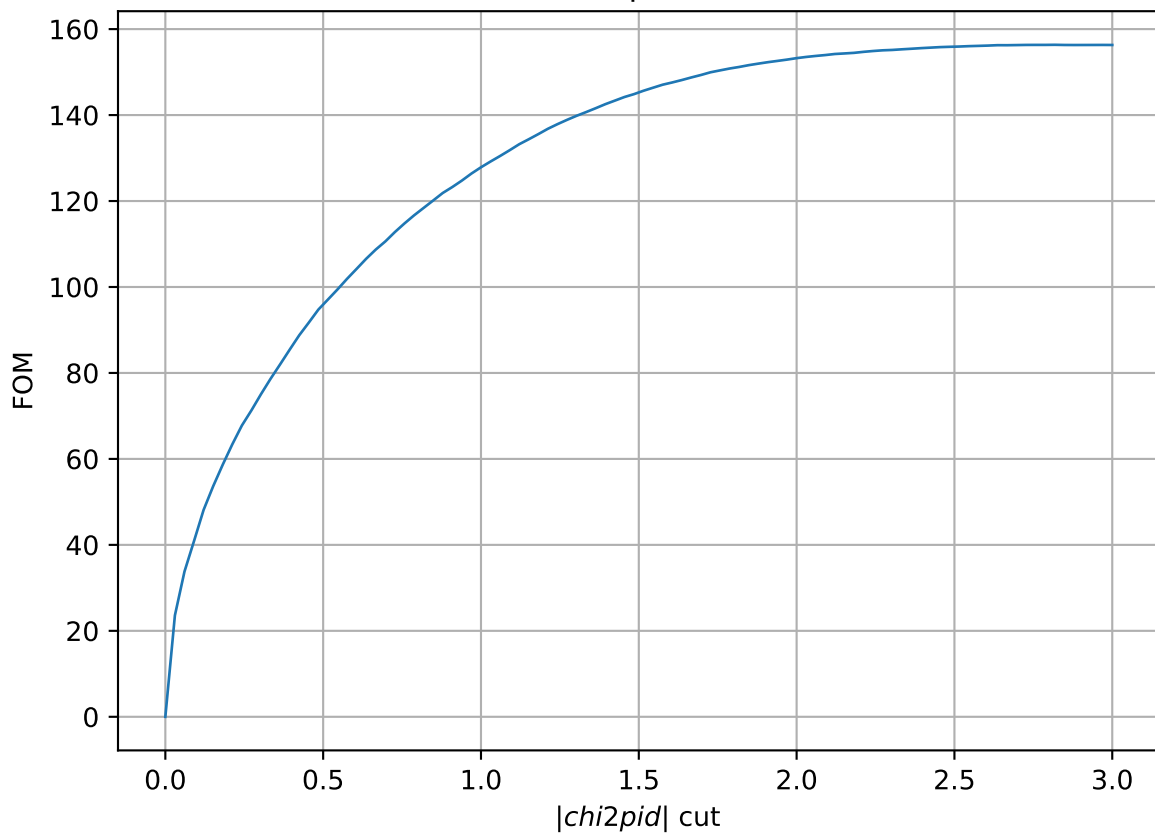
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $0.50 < p < 0.77$



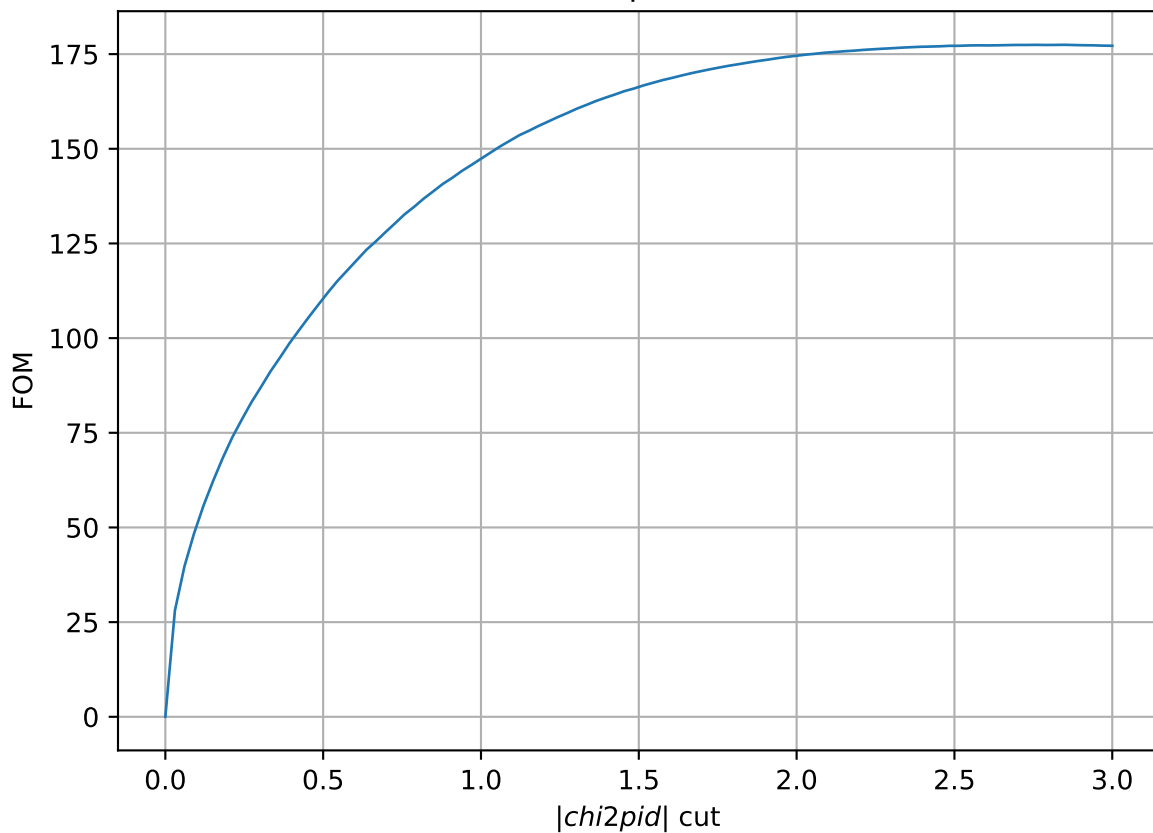
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $0.77 < p < 1.04$



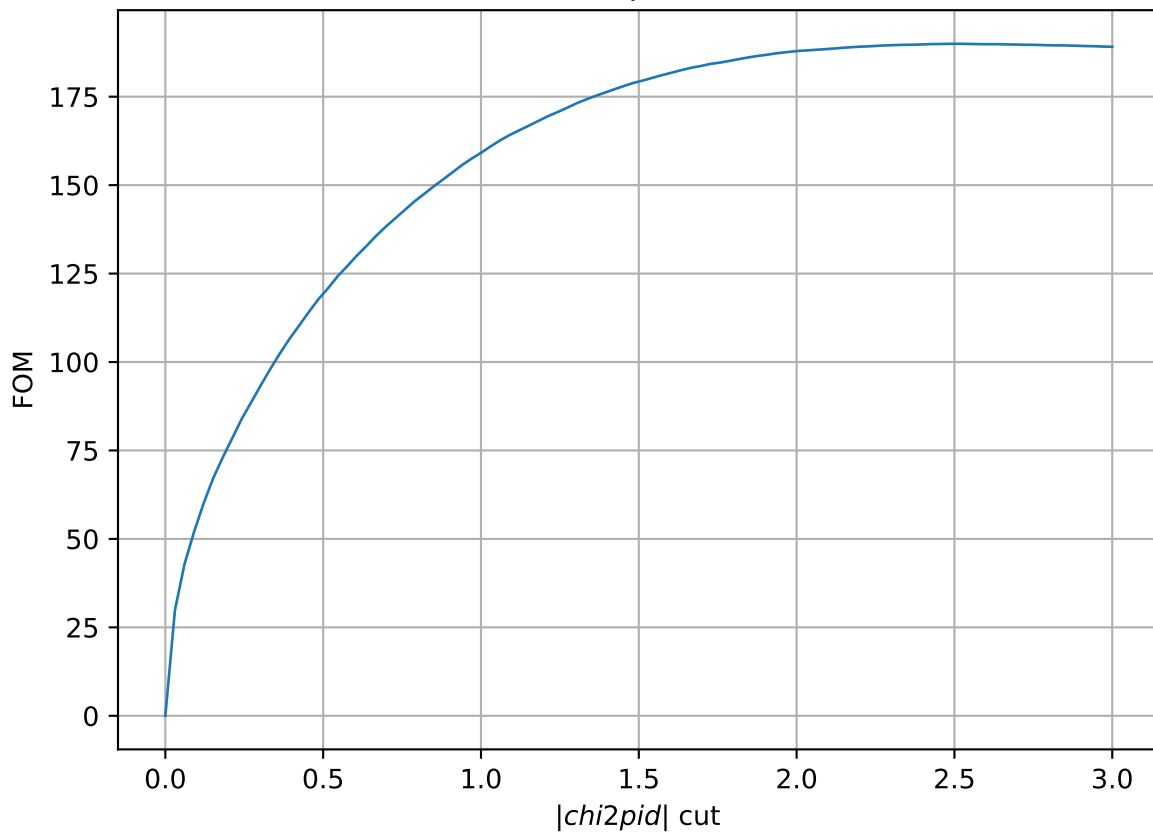
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $1.04 < p < 1.31$



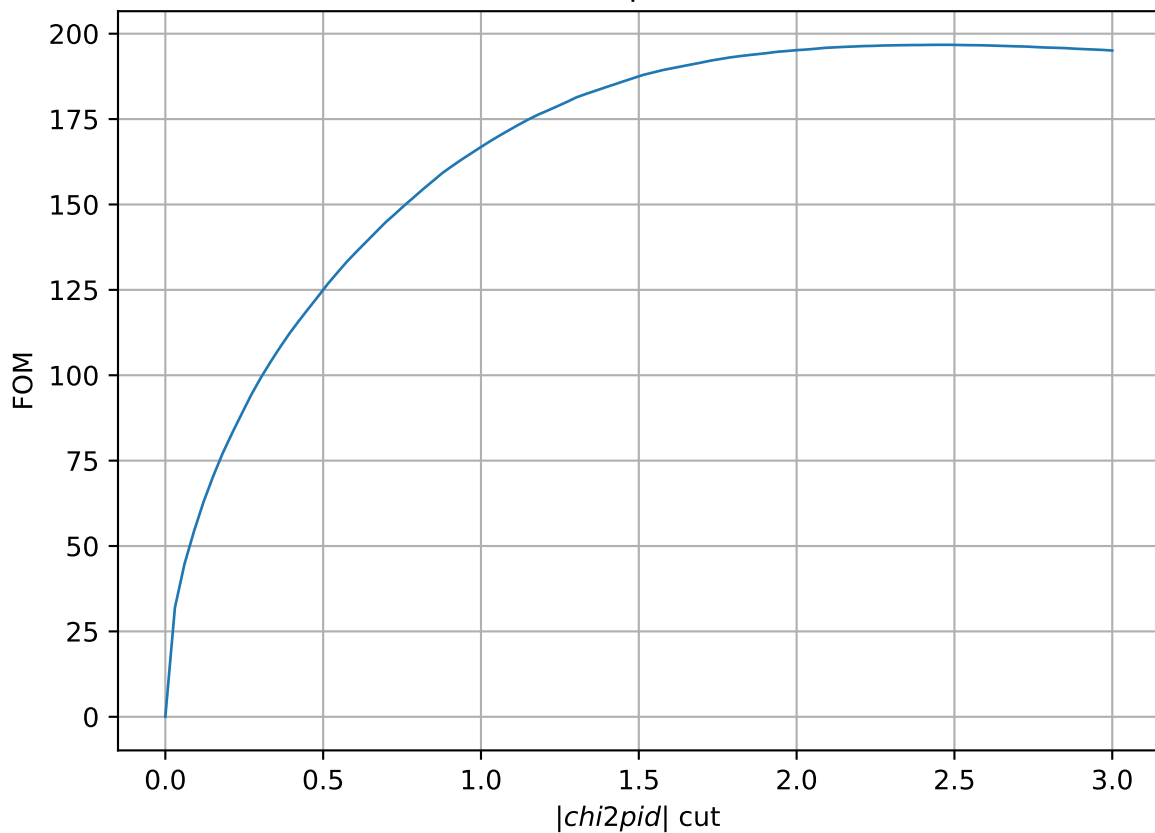
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $1.31 < p < 1.58$



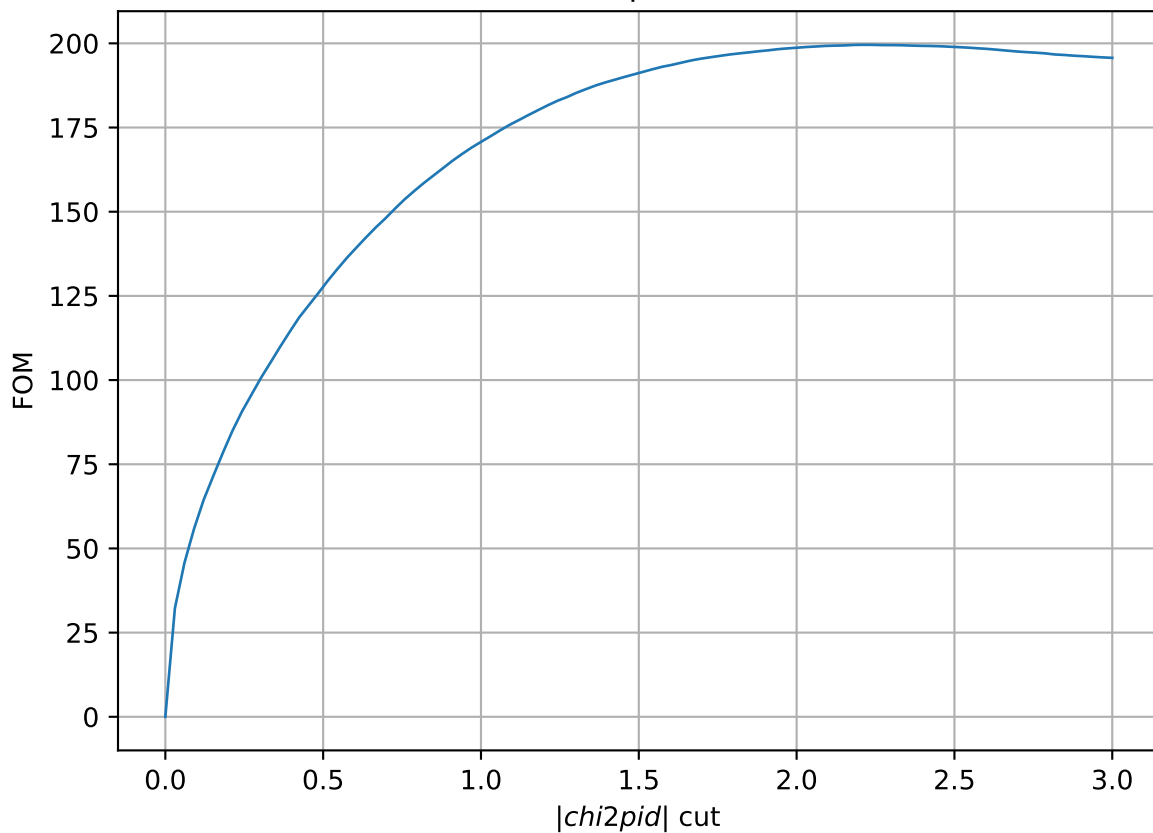
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $1.58 < p < 1.85$



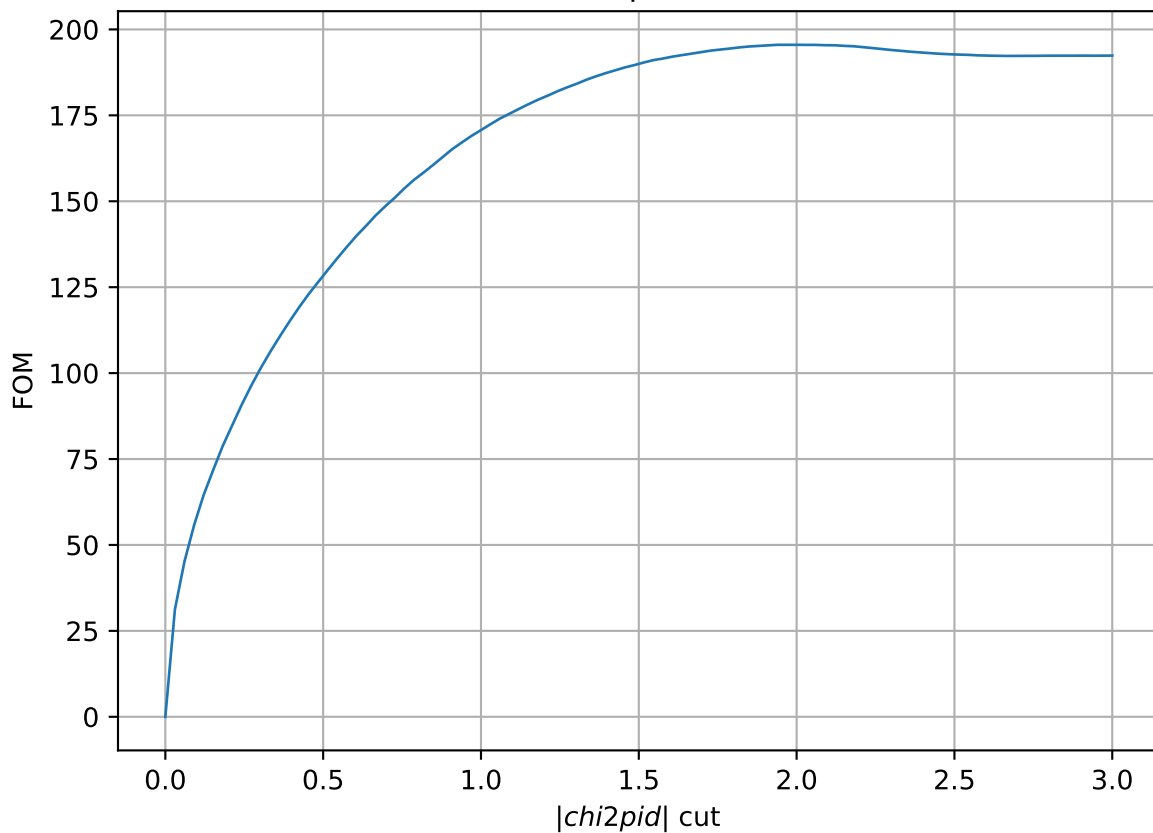
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $1.85 < p < 2.12$



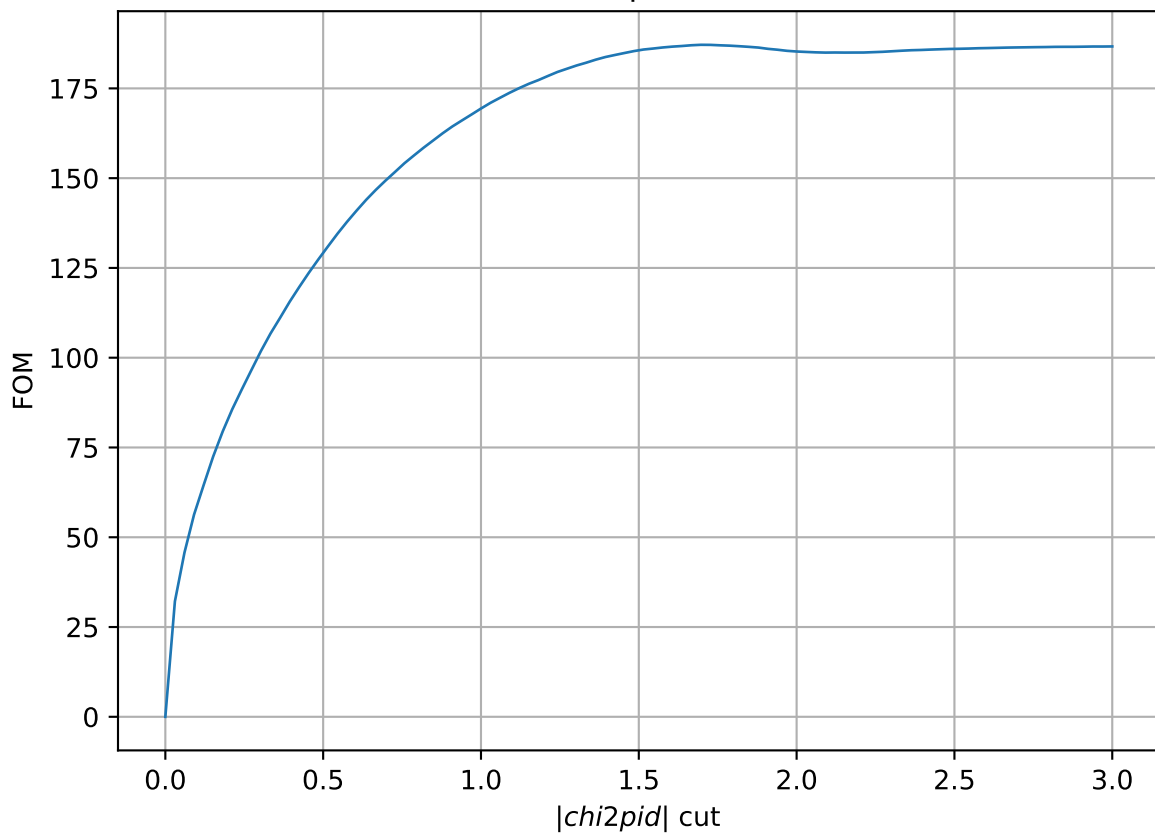
FOM vs $|\chi^2_{pid}|$ cut
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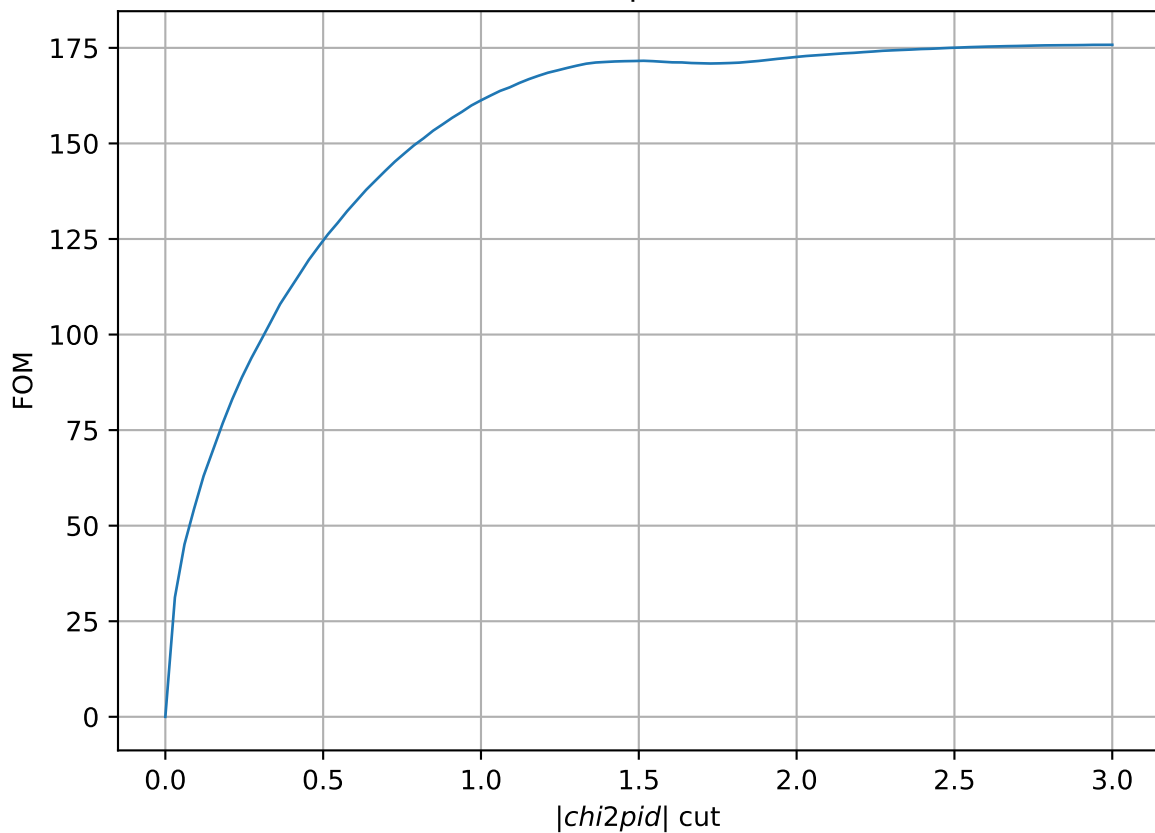
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $2.39 < p < 2.66$



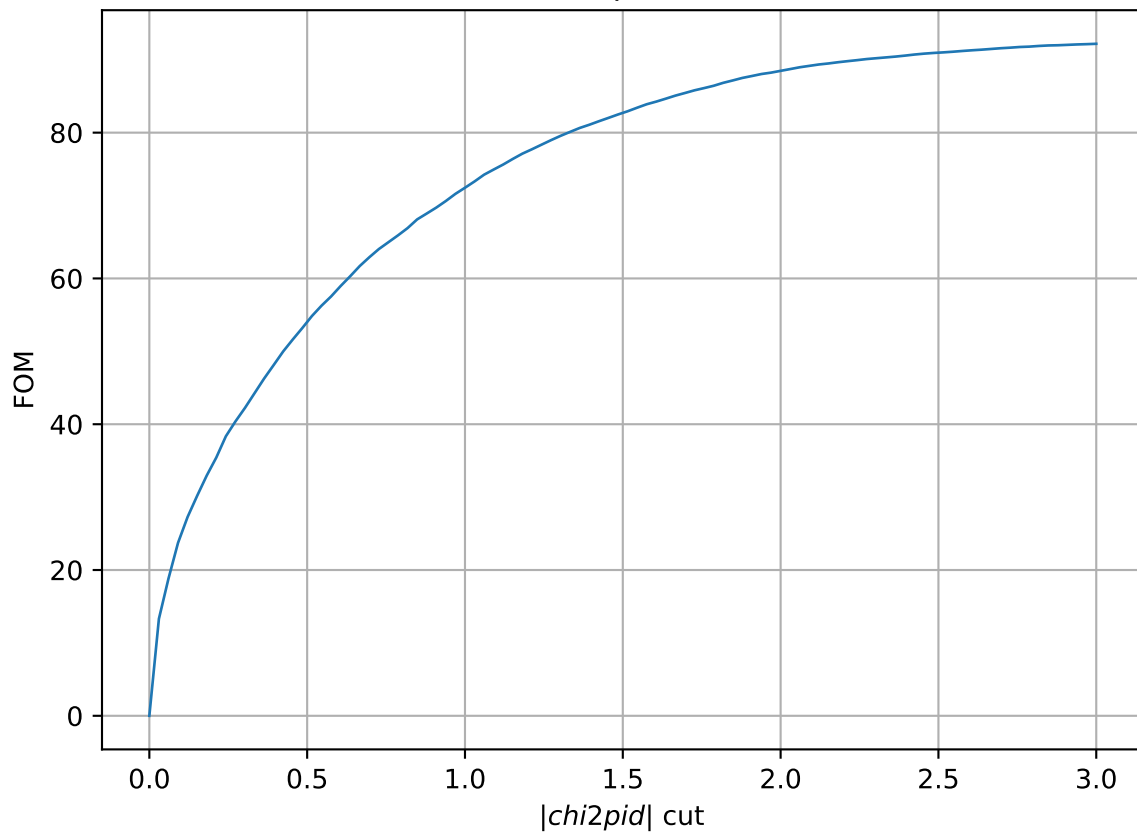
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $2.66 < p < 2.93$



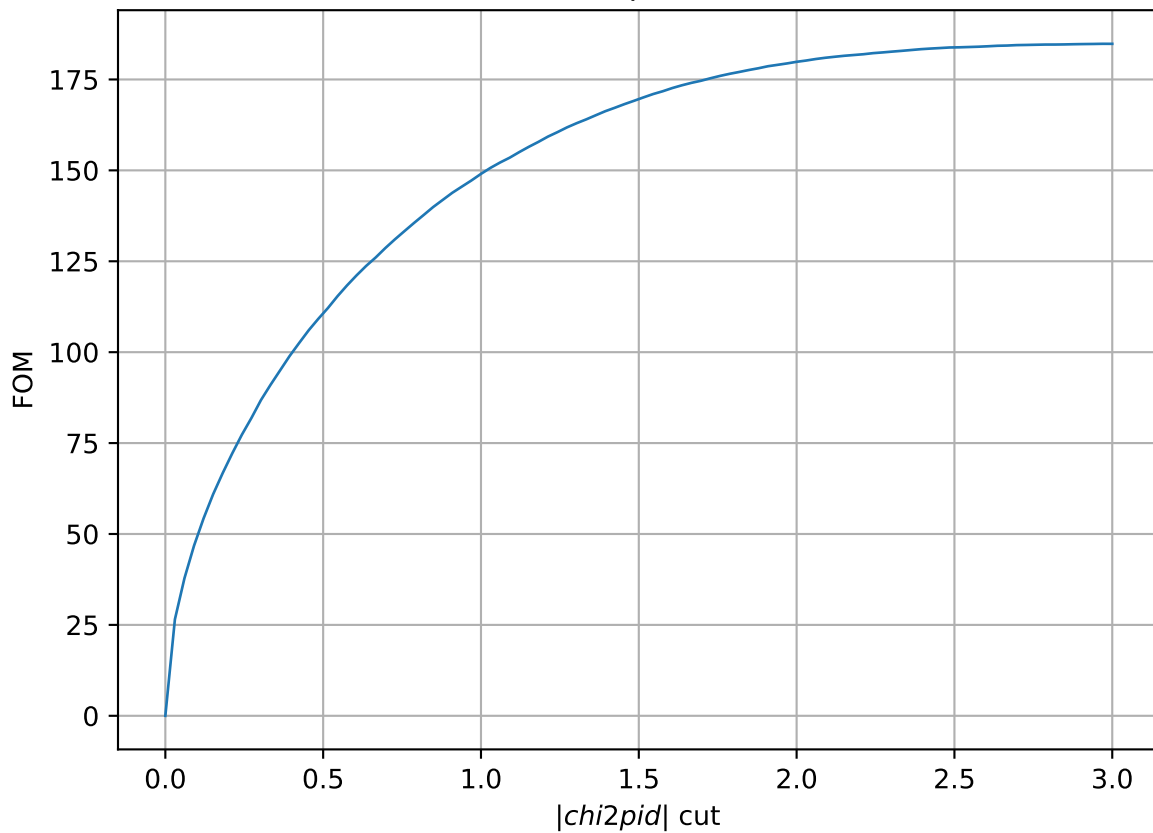
FOM vs $|\chi^2_{pid}|$ cut
 $5.0 < \theta < 11.0$
 $2.93 < p < 3.20$



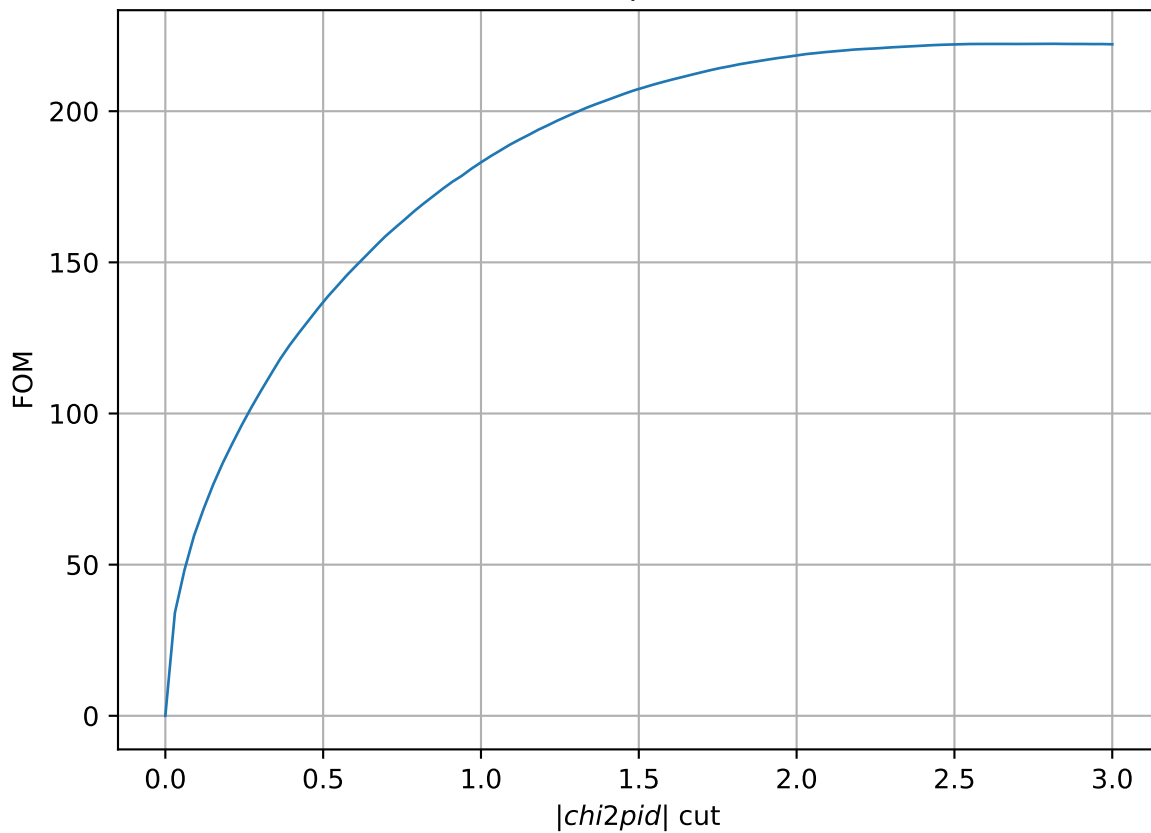
FOM vs $|\chi^2_{pid}|$ cut
 $11.0 < \theta < 17.0$
 $0.50 < p < 0.77$



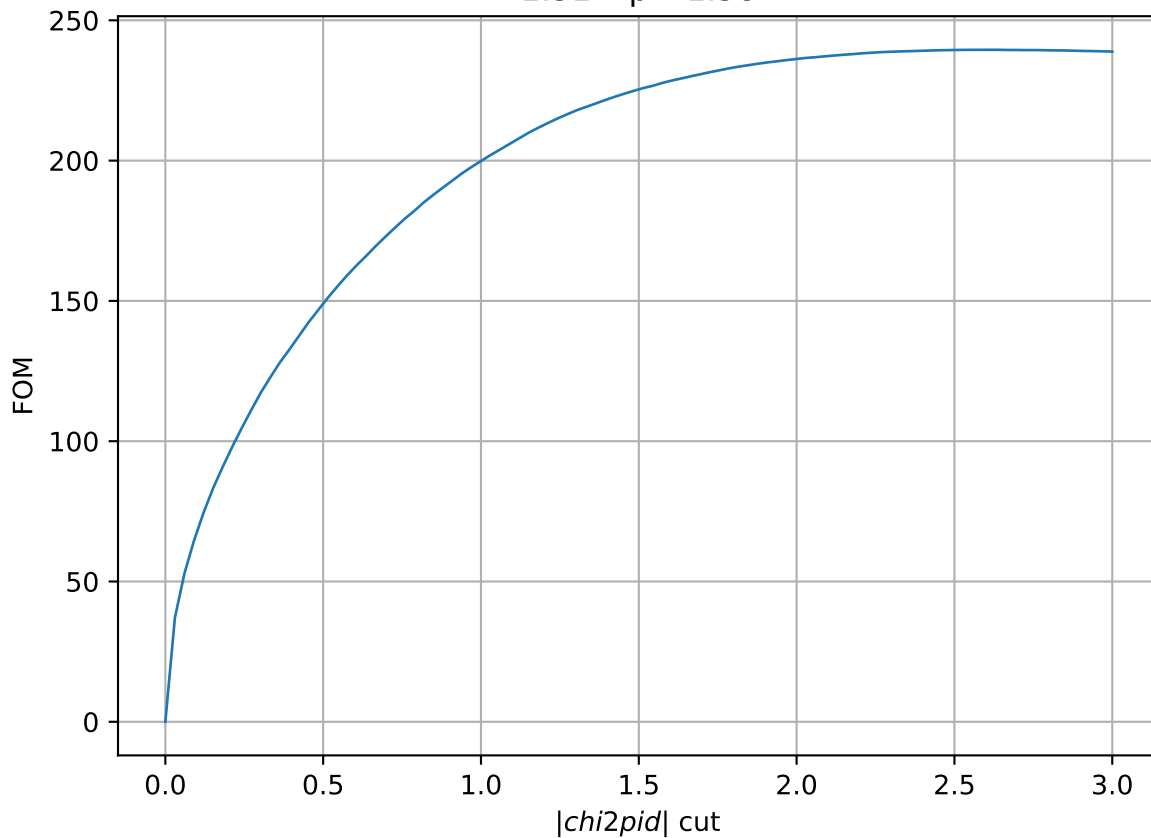
FOM vs $|\chi^2_{pid}|$ cut
 $11.0 < \theta < 17.0$
 $0.77 < p < 1.04$



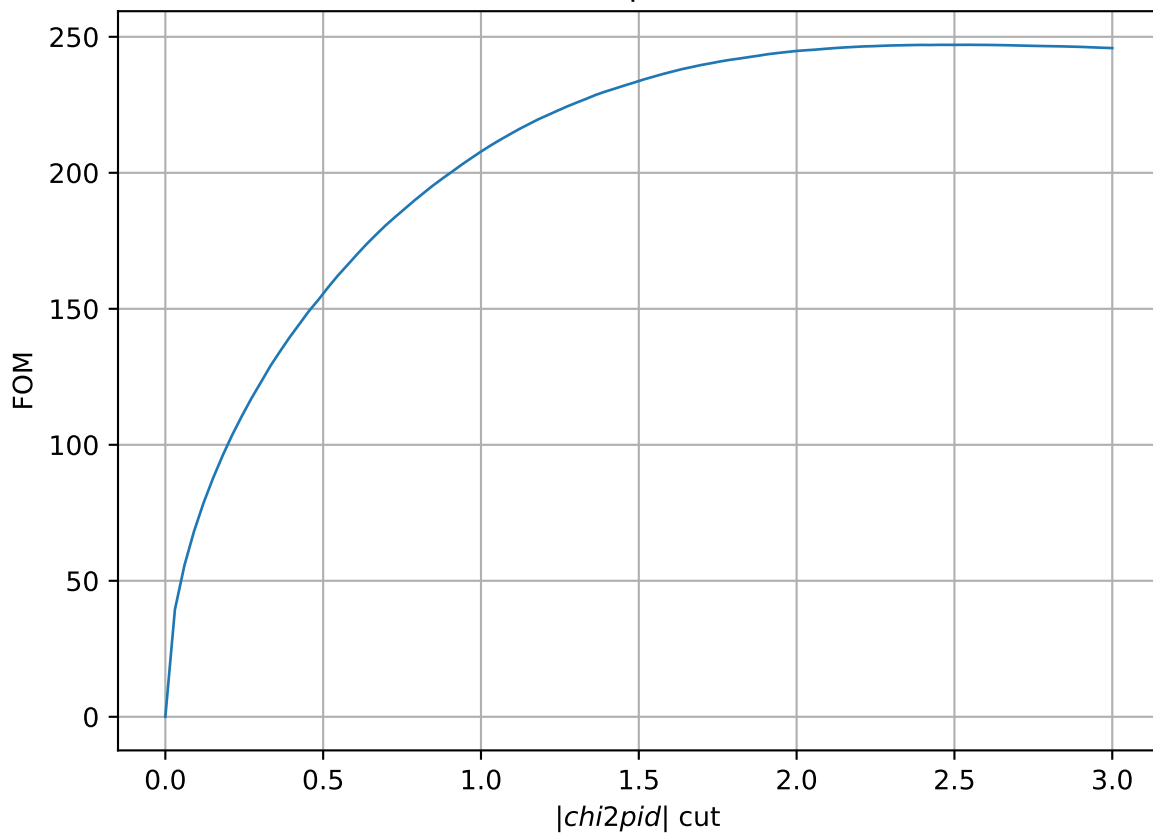
FOM vs $|\chi^2_{pid}|$ cut
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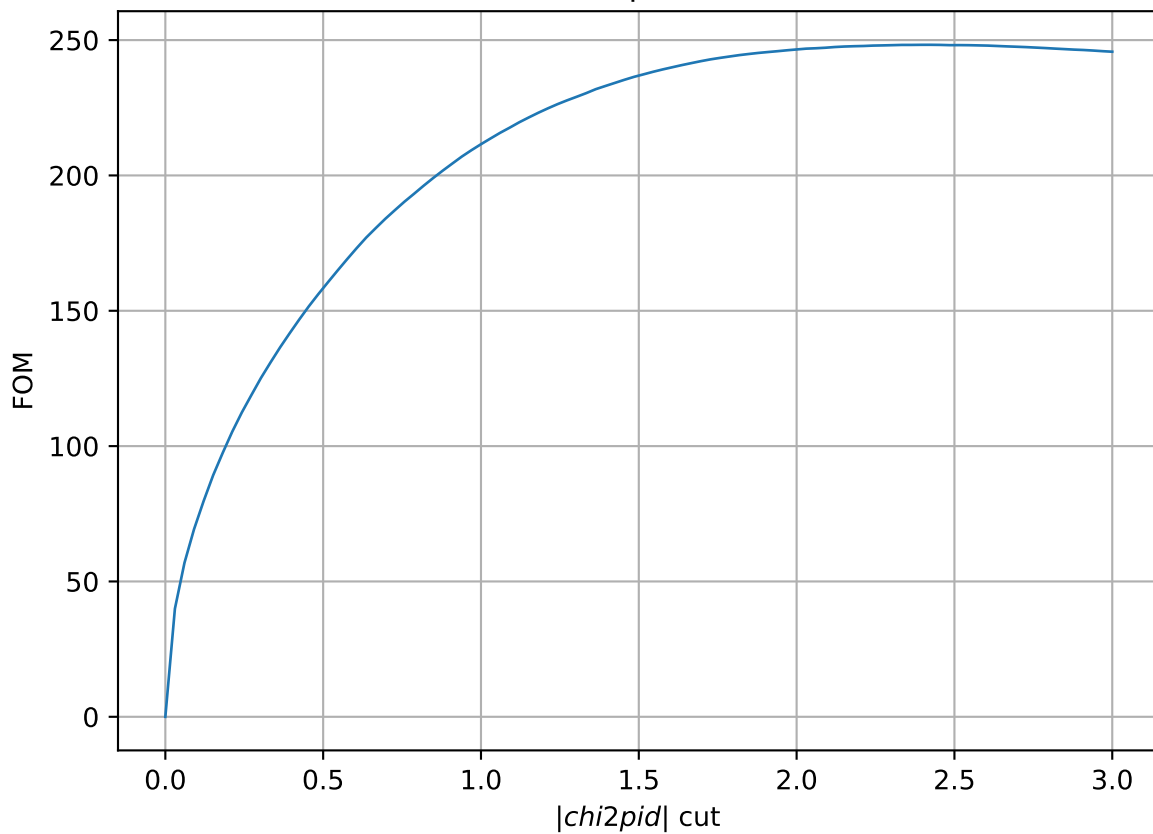
FOM vs $|\chi^2_{pid}|$ cut
 $11.0 < \theta < 17.0$
 $1.31 < p < 1.58$



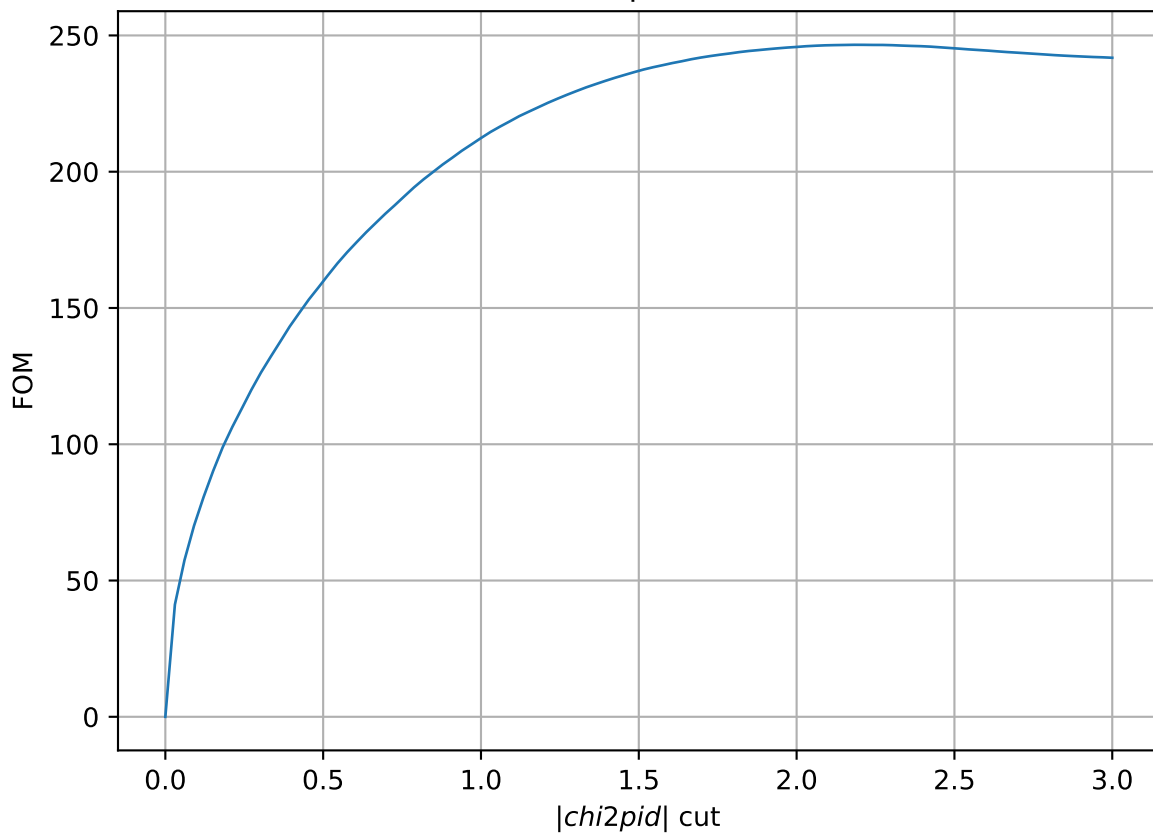
FOM vs $|\chi^2_{pid}|$ cut
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 $1.58 < p < 1.85$



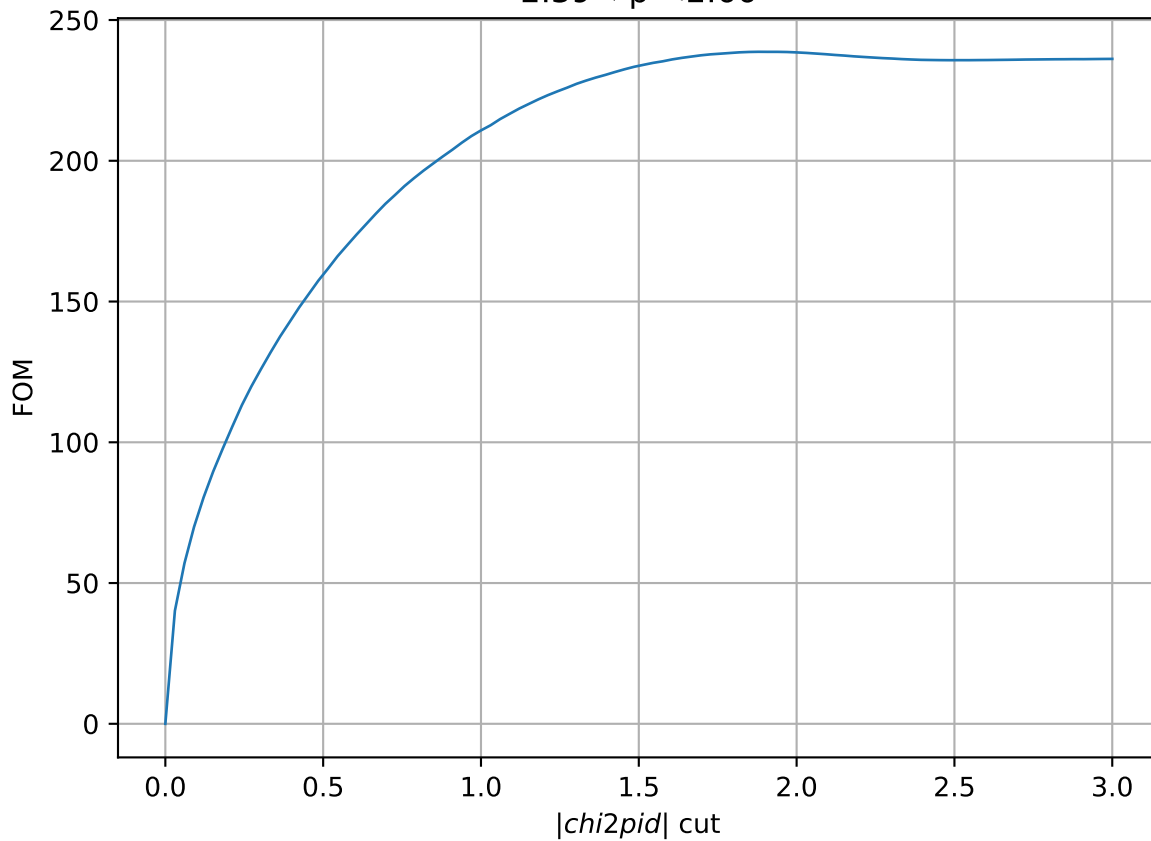
FOM vs $|\chi^2_{pid}|$ cut
 $11.0 < \theta < 17.0$
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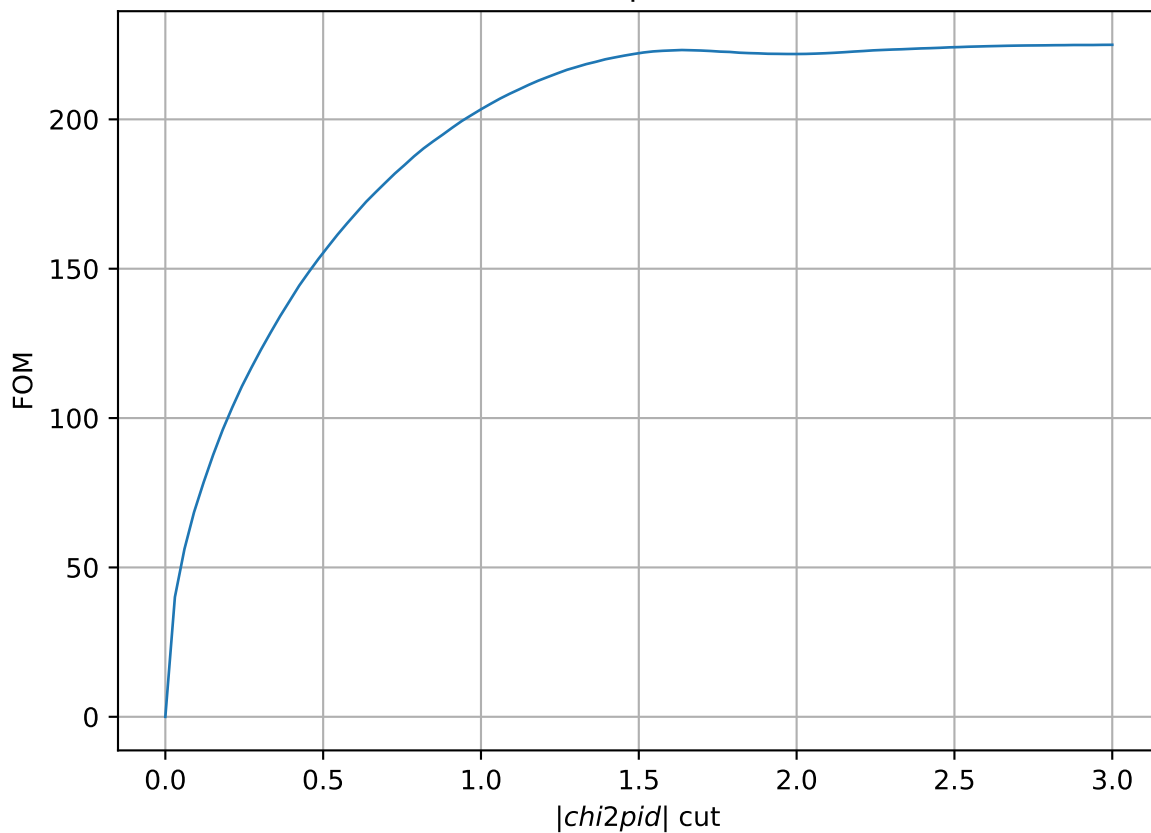
FOM vs $|\chi^2_{pid}|$ cut
 $11.0 < \theta < 17.0$
 $2.12 < p < 2.39$



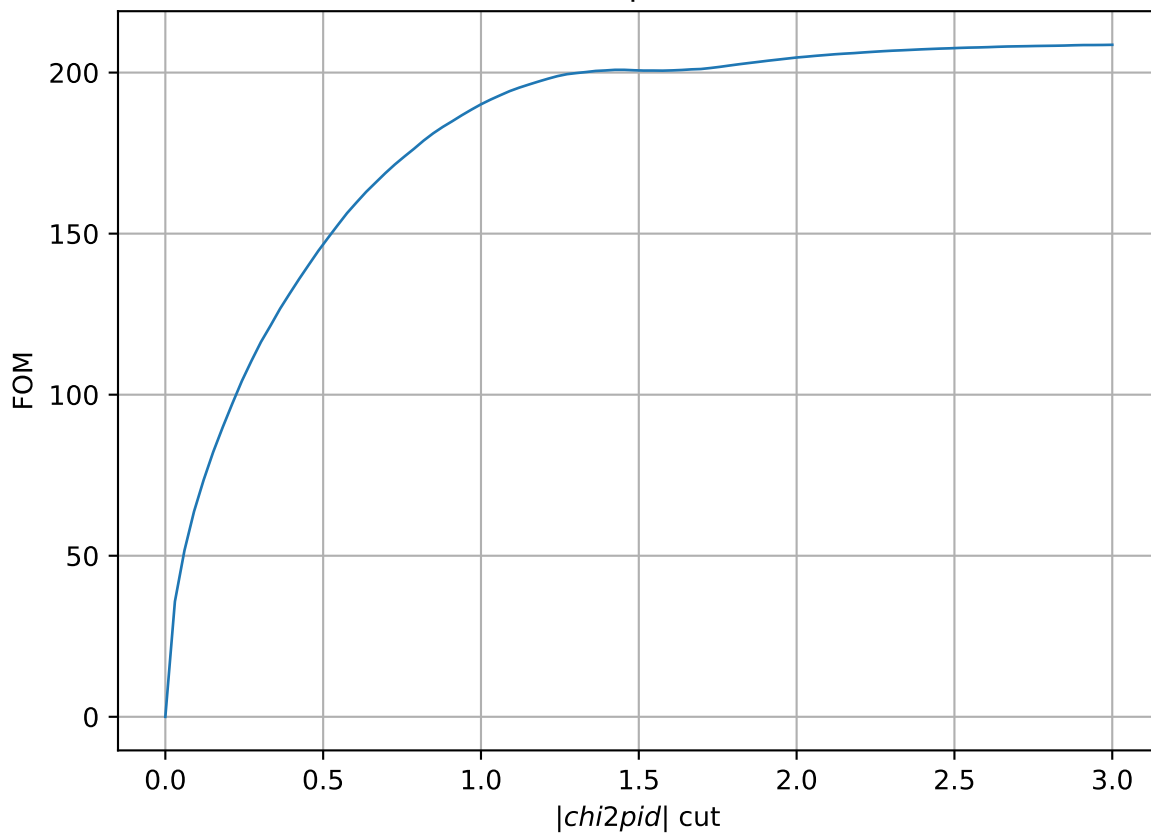
FOM vs $|\chi^2_{pid}|$ cut
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 $2.39 < p < 2.66$



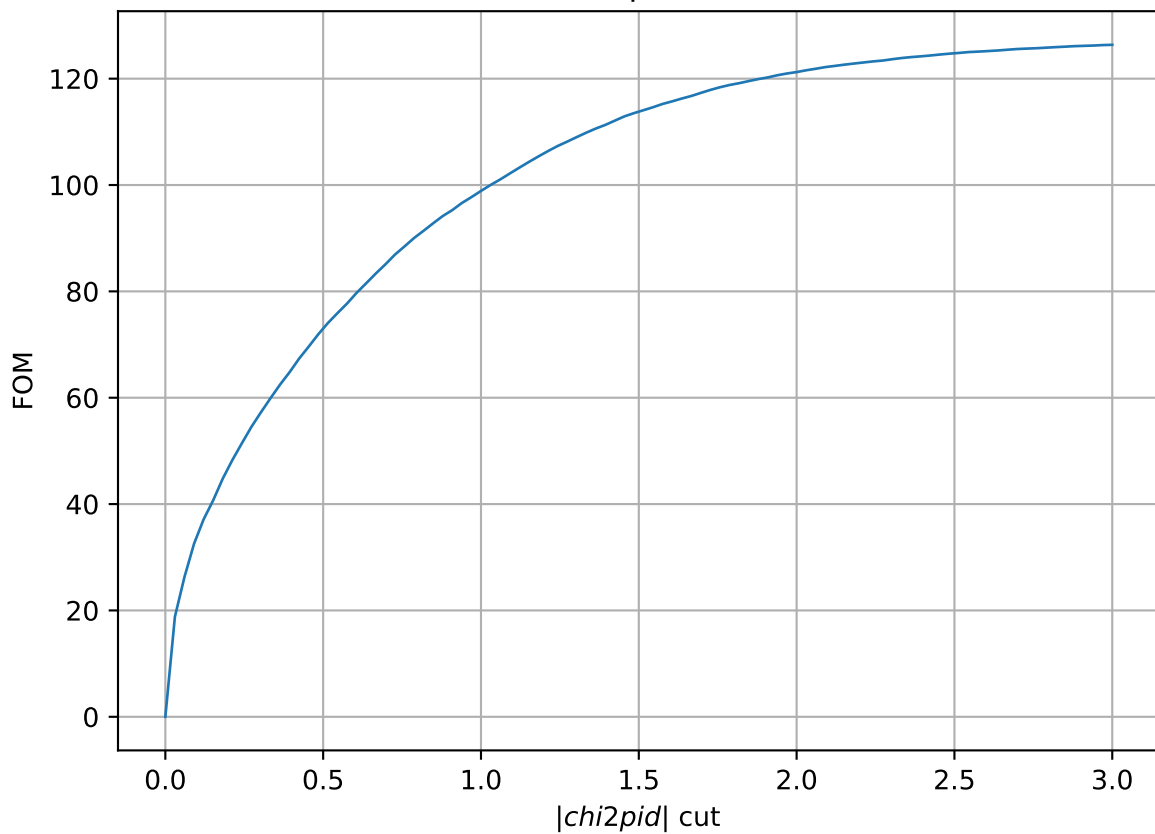
FOM vs $|\chi^2_{pid}|$ cut
 $11.0 < \theta < 17.0$
 $2.66 < p < 2.93$



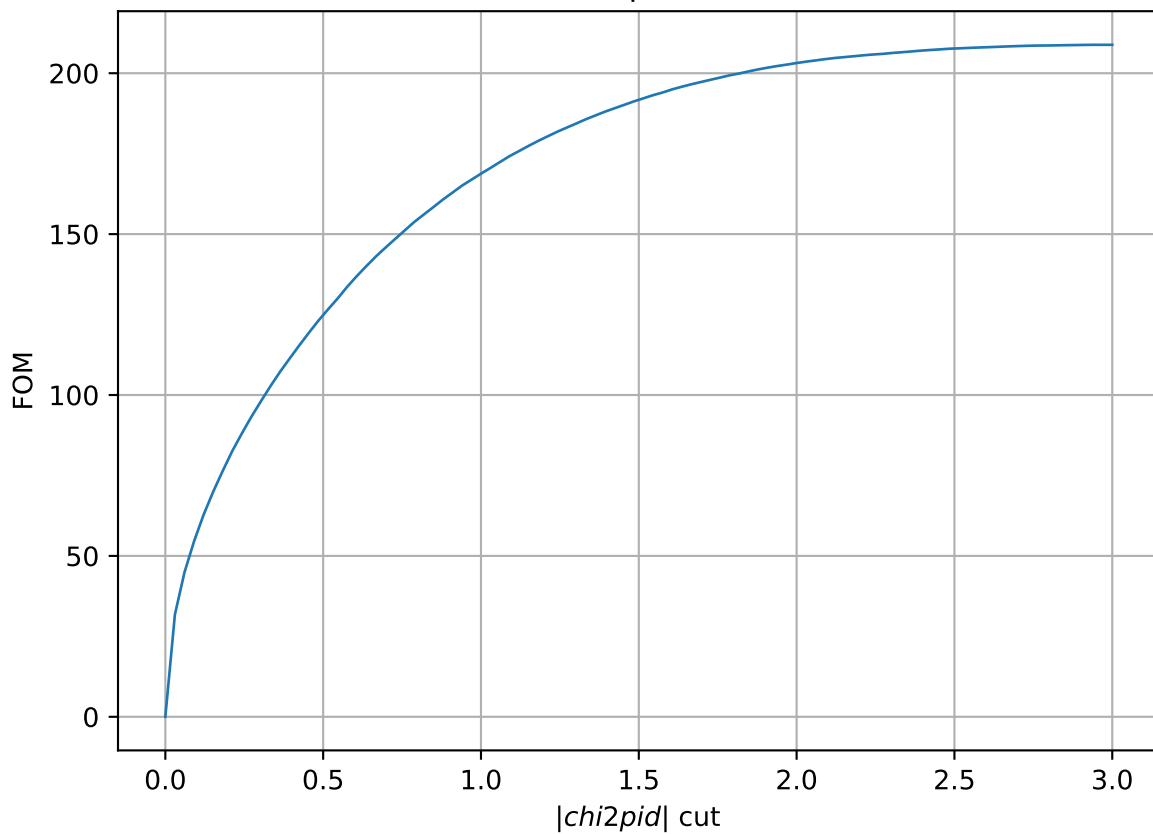
FOM vs $|\chi^2_{pid}|$ cut
 $11.0 < \theta < 17.0$
 $2.93 < p < 3.20$



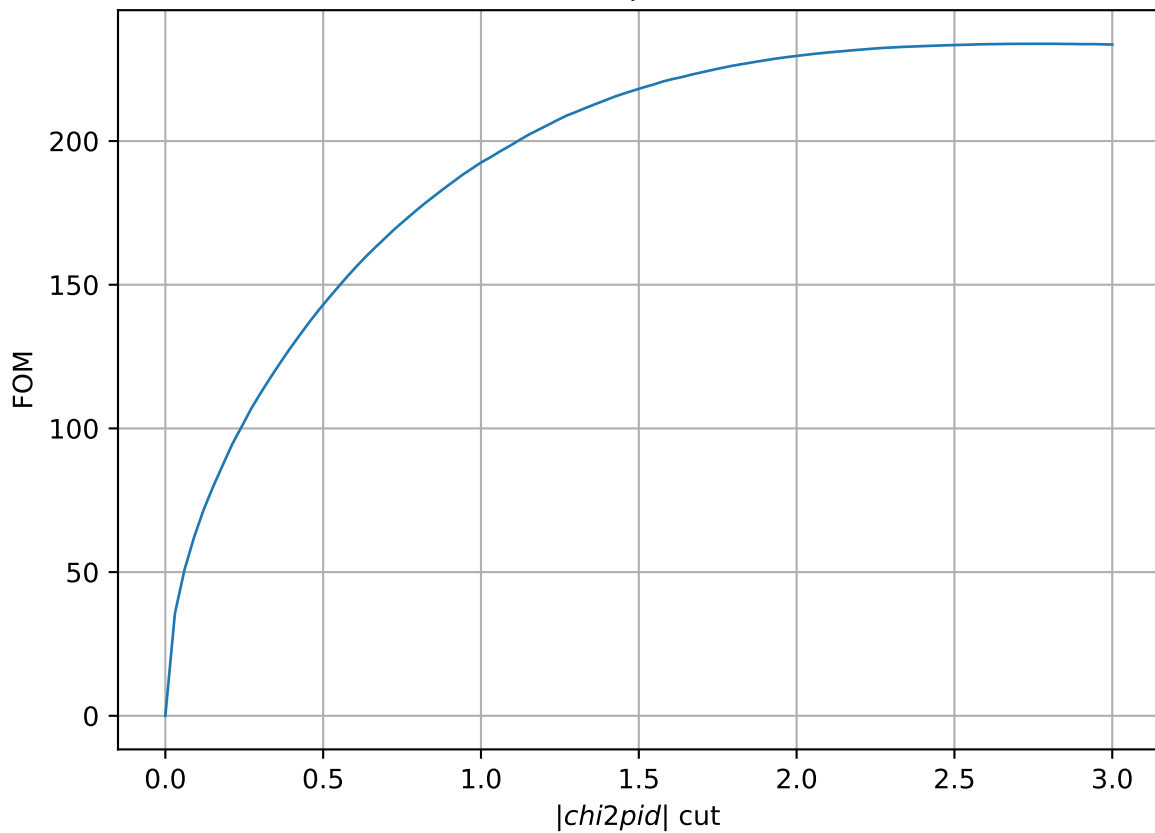
FOM vs $|\chi^2_{pid}|$ cut
 $17.0 < \theta < 23.0$
 $0.50 < p < 0.77$



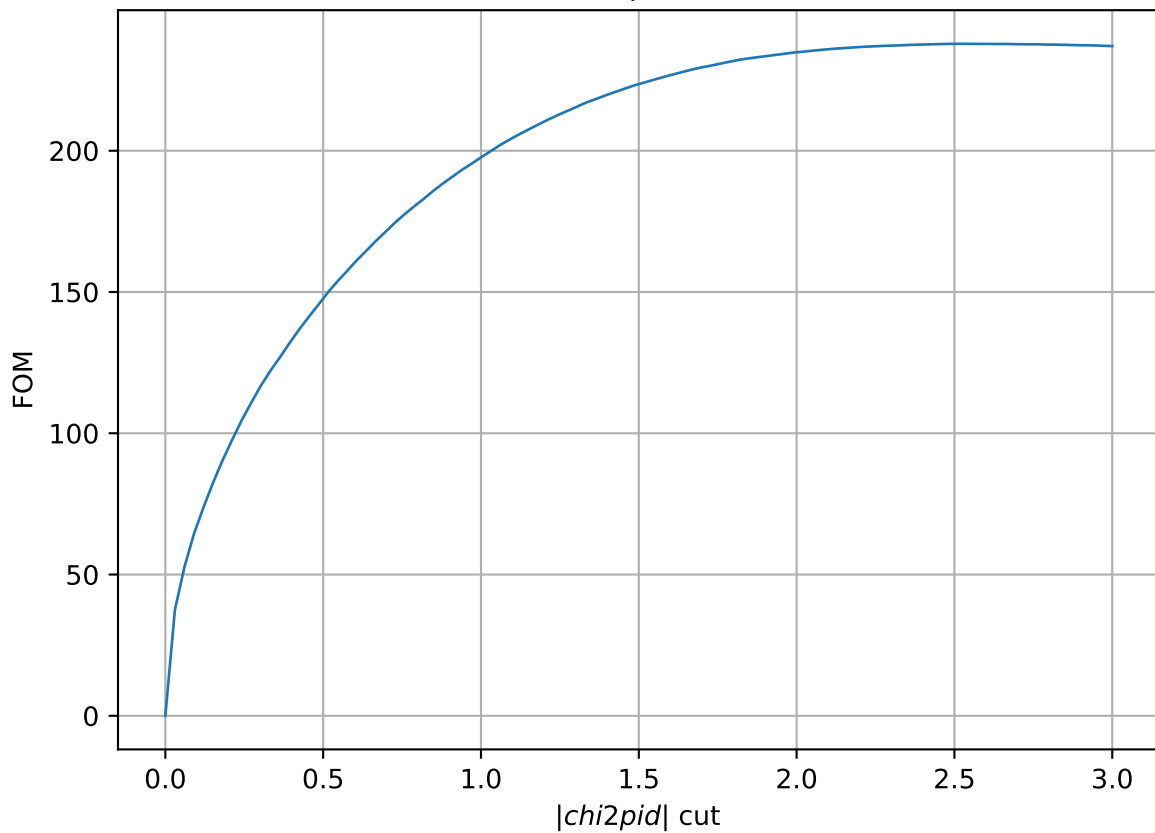
FOM vs $|\chi^2_{pid}|$ cut
 $17.0 < \theta < 23.0$
 $0.77 < p < 1.04$



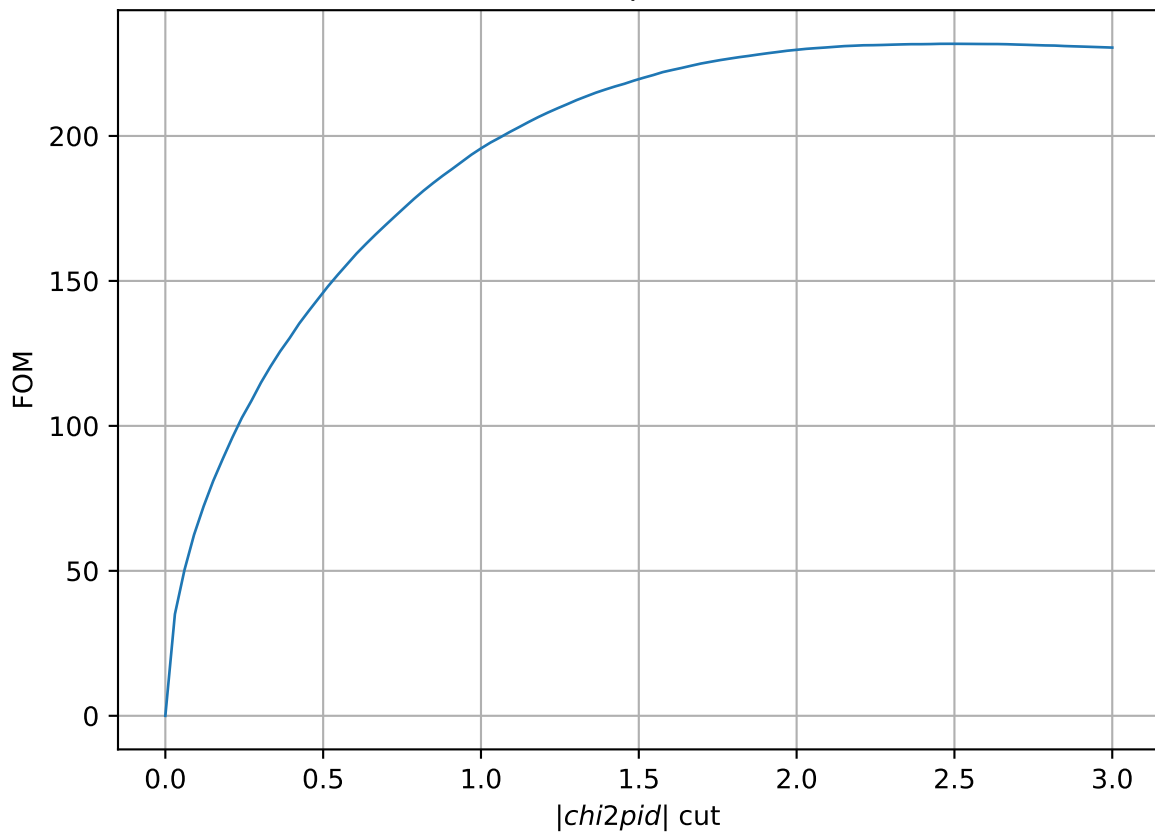
FOM vs $|\chi^2_{pid}|$ cut
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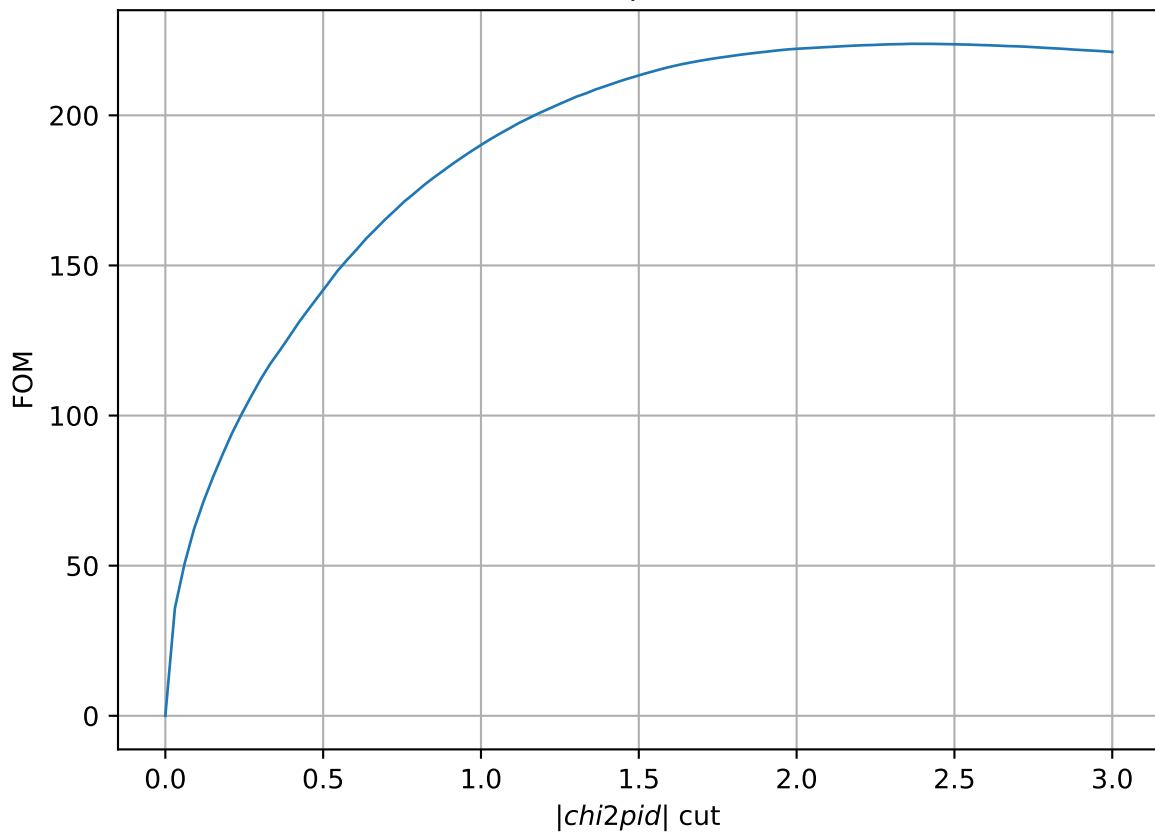
FOM vs $|\chi^2_{pid}|$ cut
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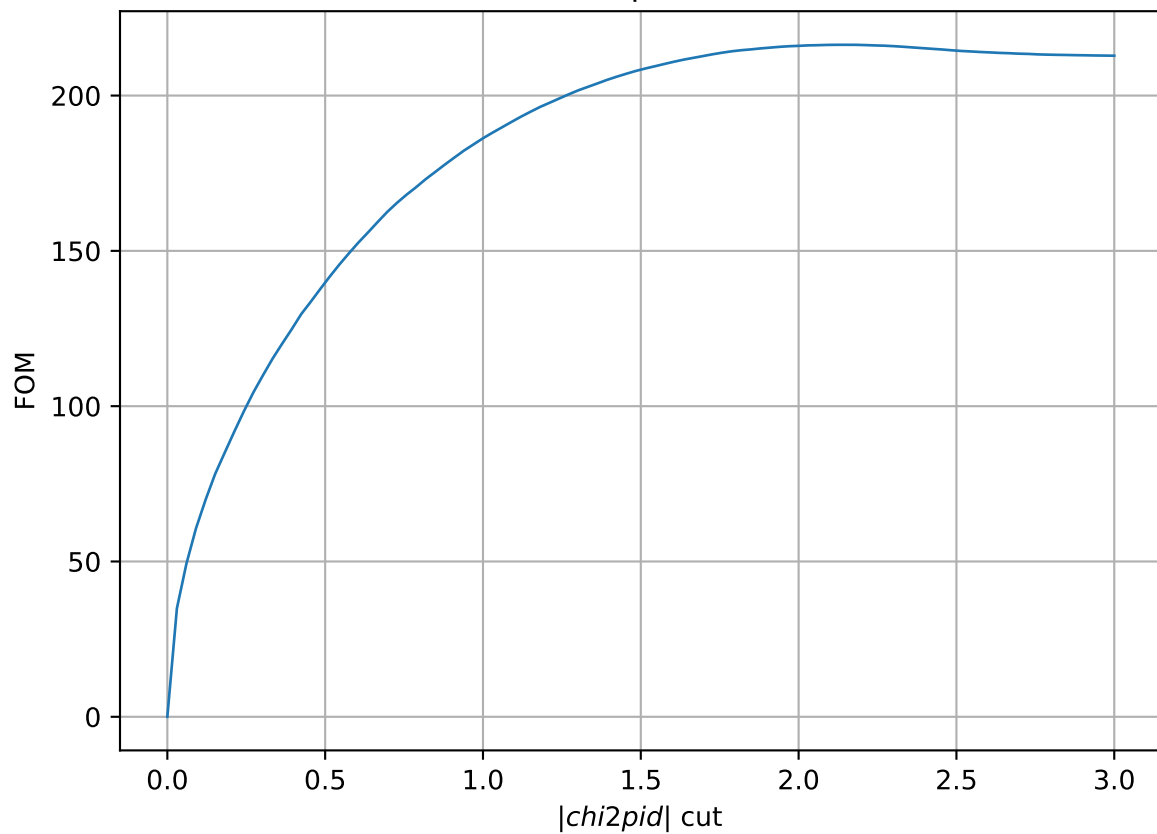
FOM vs $|\chi^2_{pid}|$ cut
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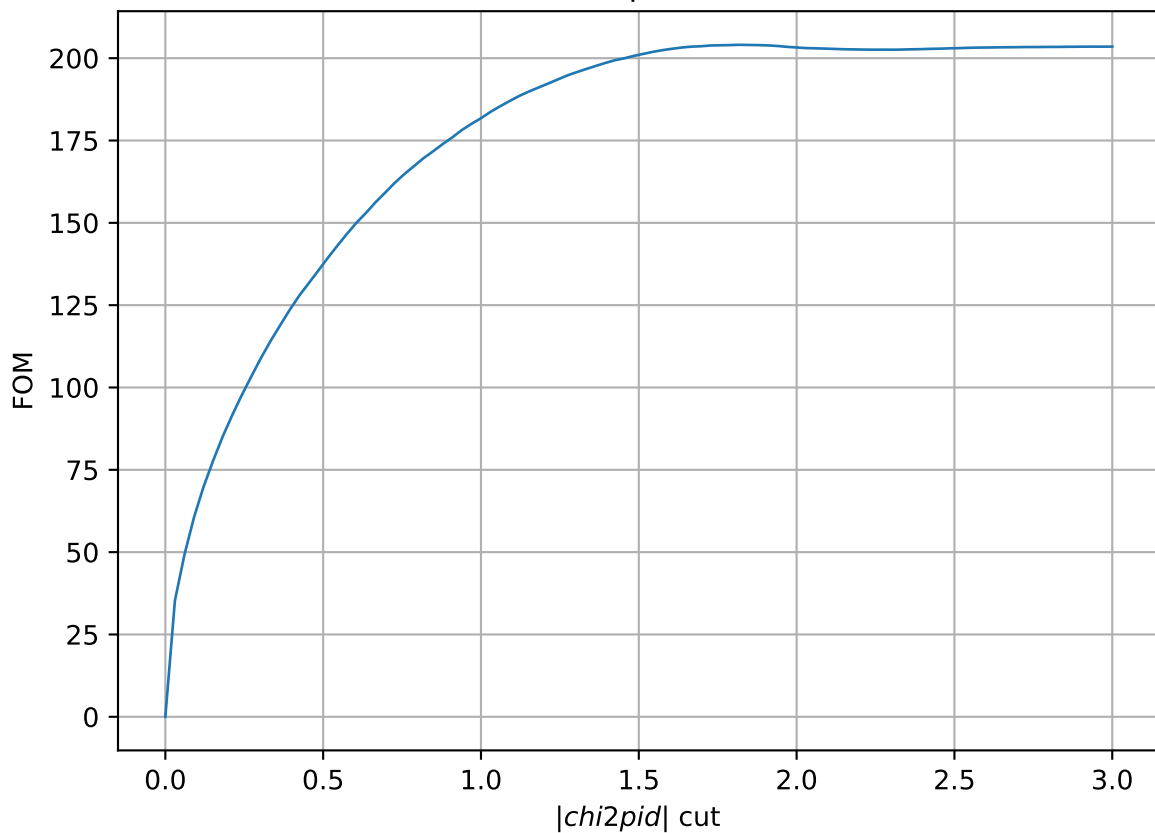
FOM vs $|\chi^2_{pid}|$ cut
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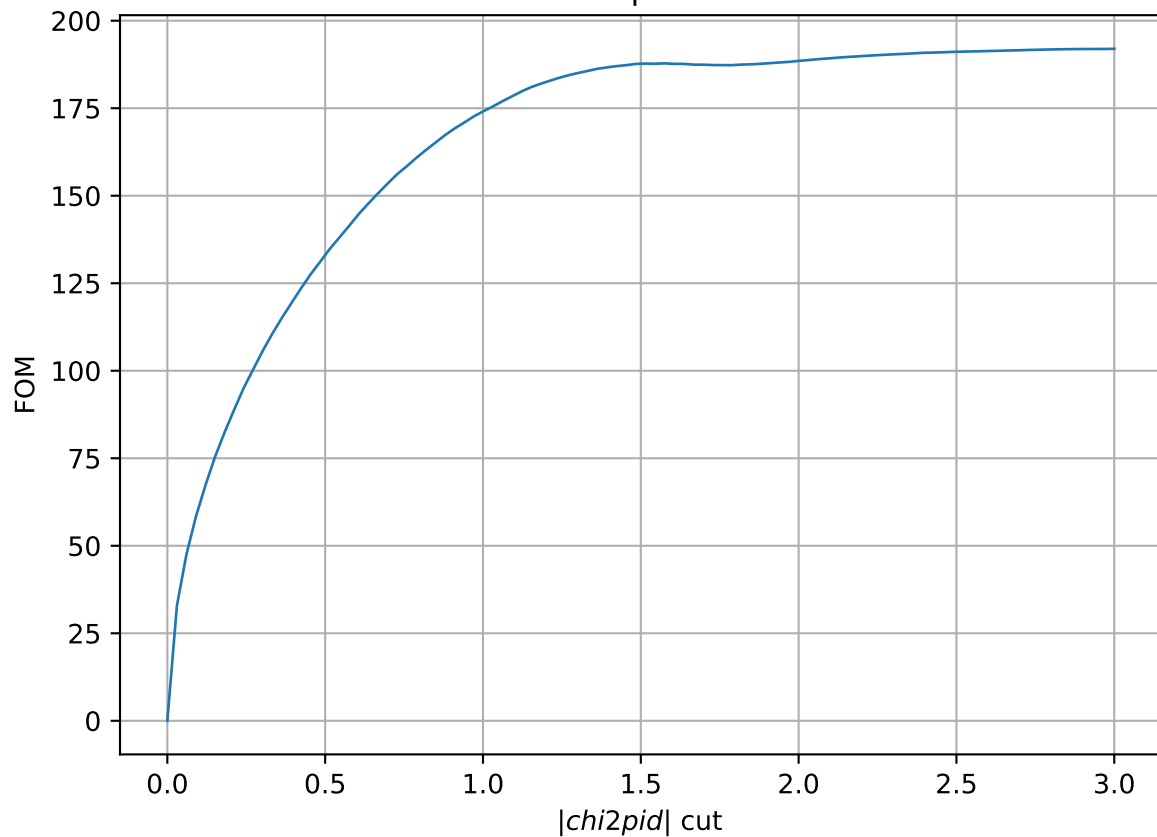
FOM vs $|\chi^2_{pid}|$ cut
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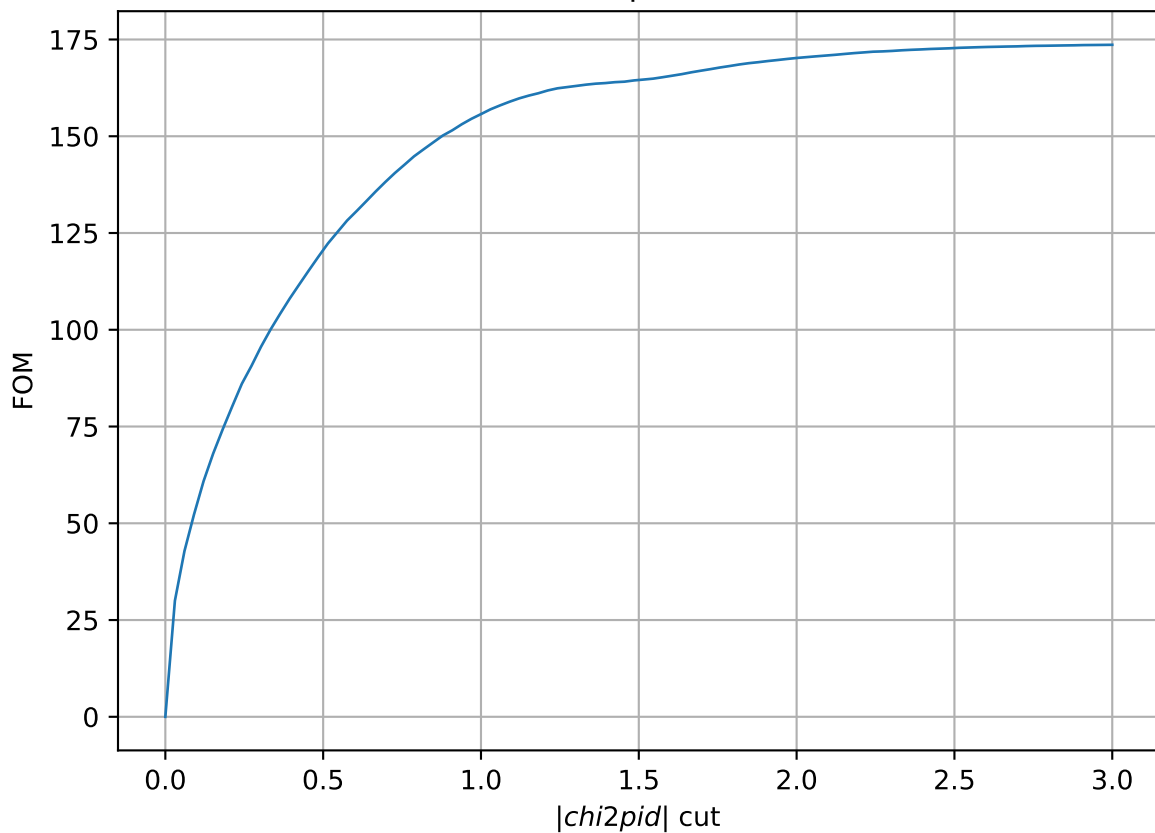
FOM vs $|\chi^2_{pid}|$ cut
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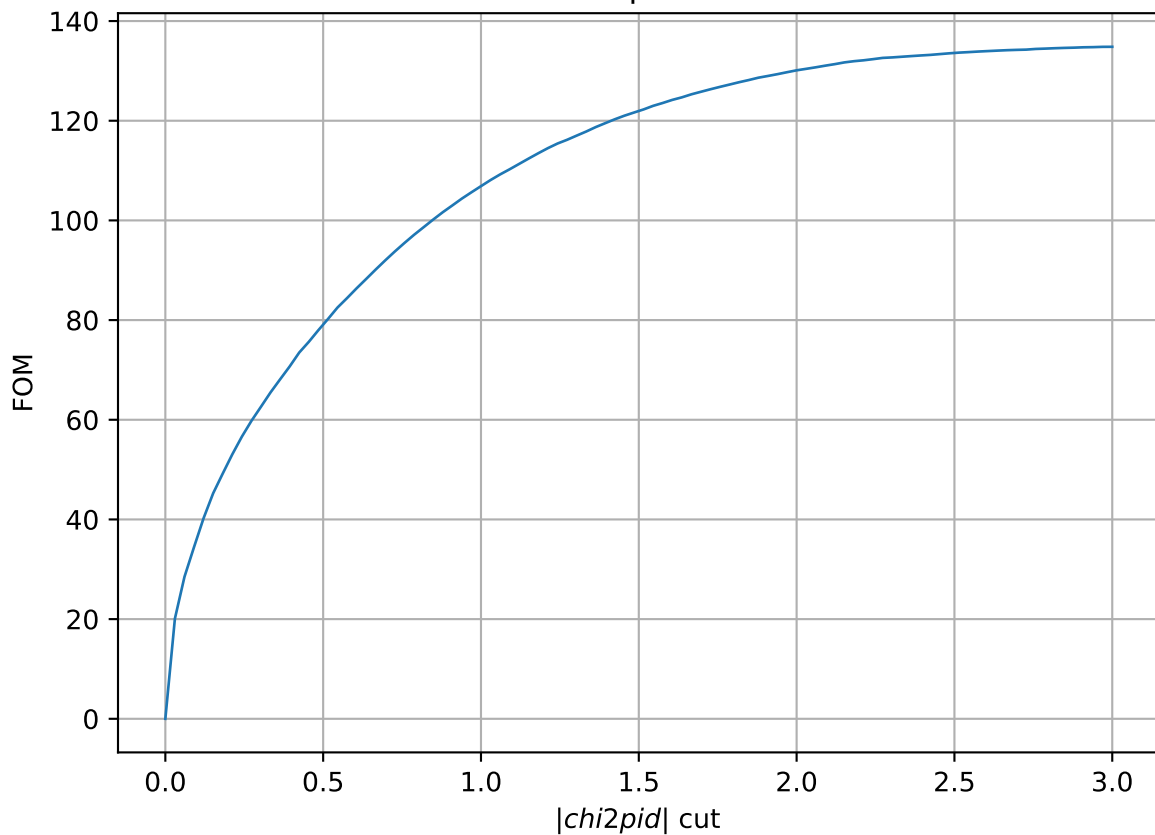
FOM vs $|\chi^2_{pid}|$ cut
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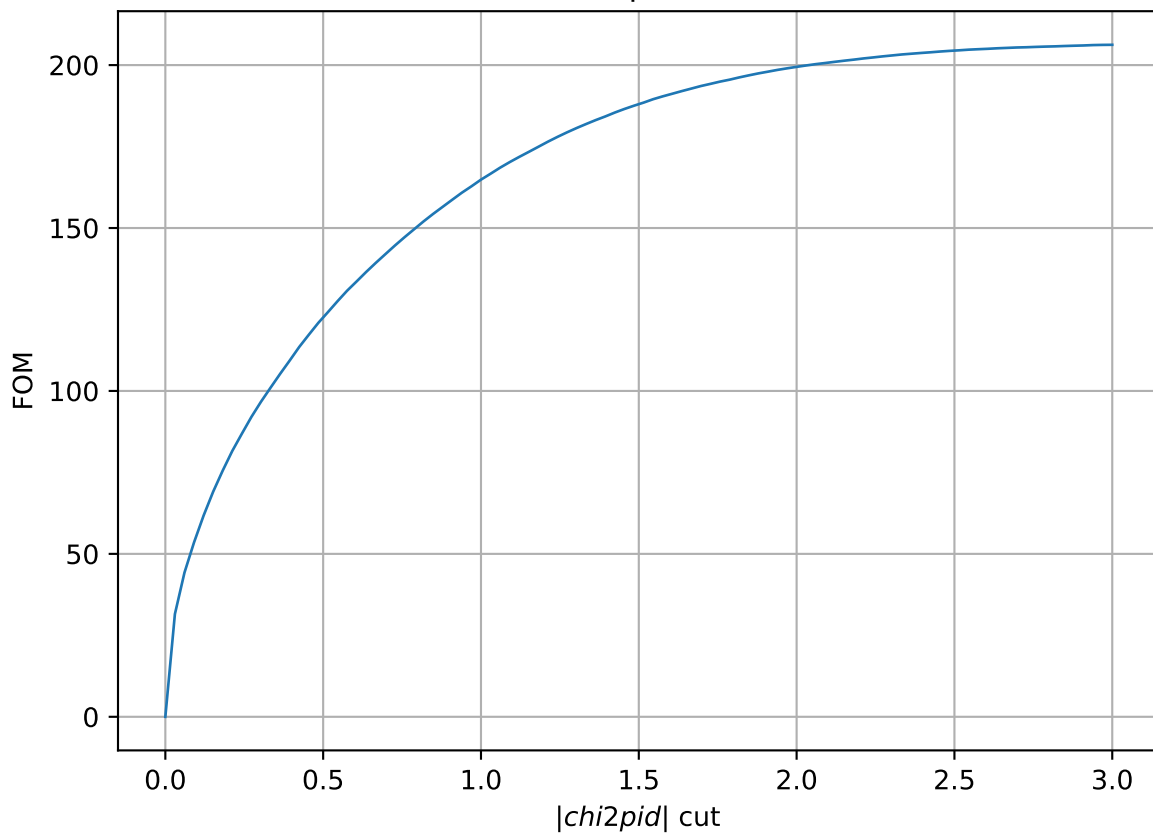
FOM vs $|\chi^2_{pid}|$ cut
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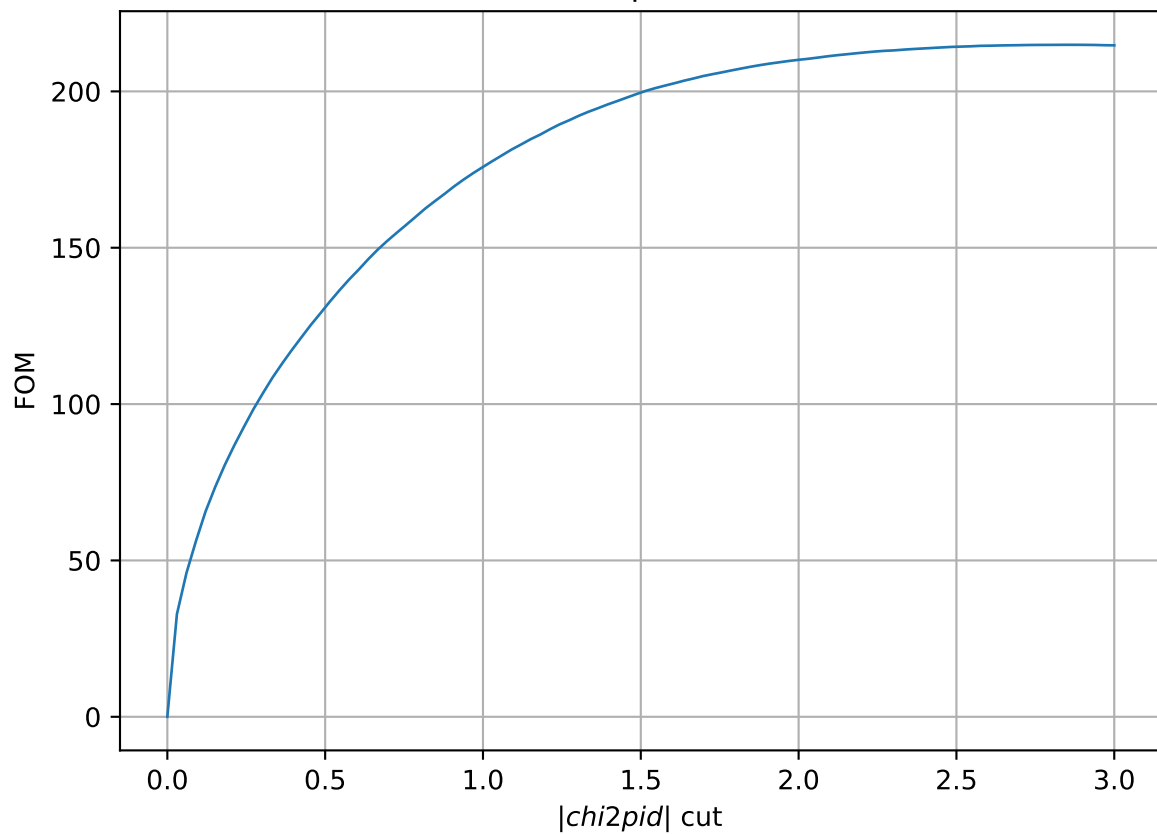
FOM vs $|\chi^2_{pid}|$ cut
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 $0.50 < p < 0.77$



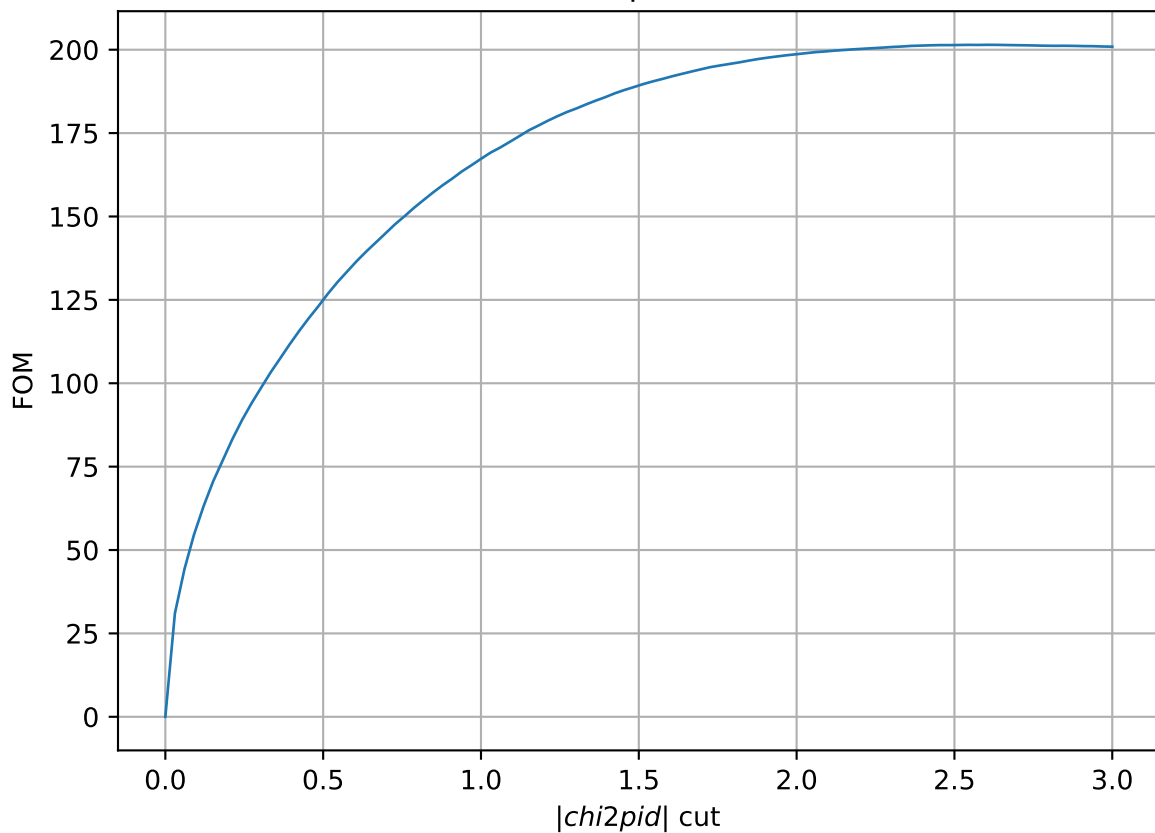
FOM vs $|\chi^2_{pid}|$ cut
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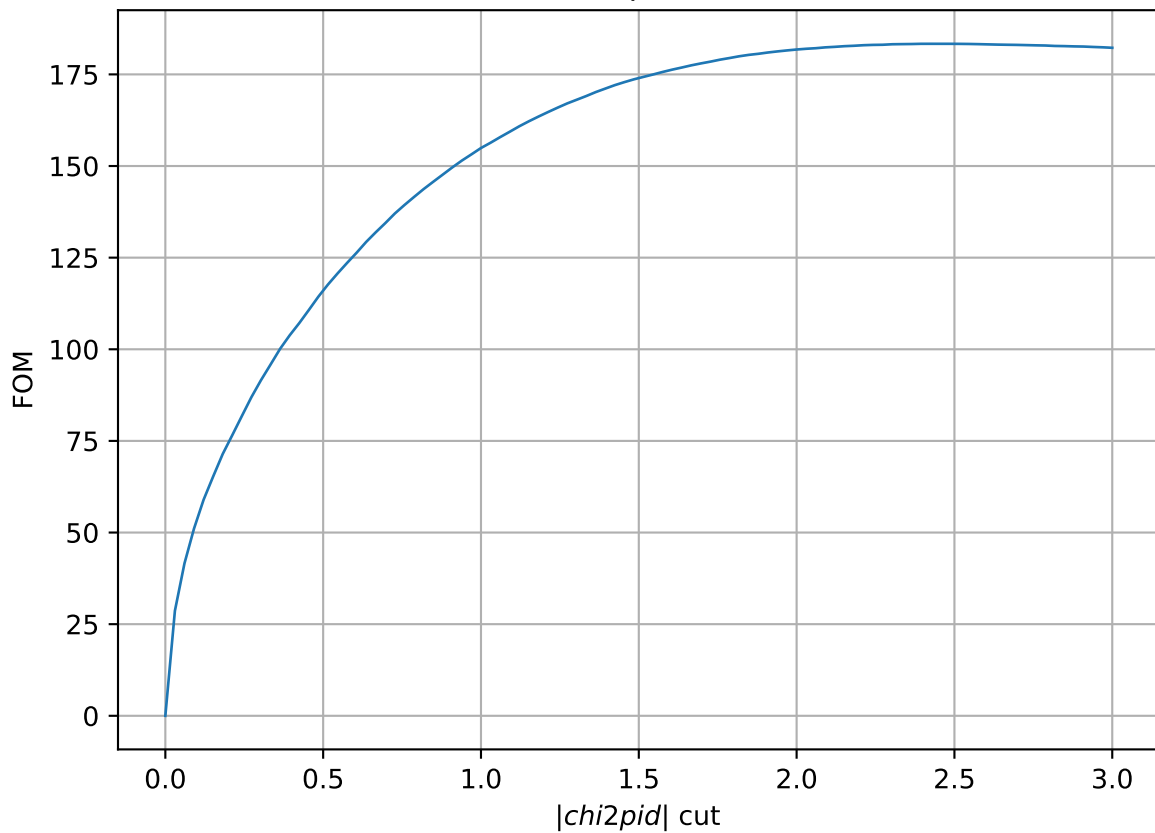
FOM vs $|\chi^2_{pid}|$ cut
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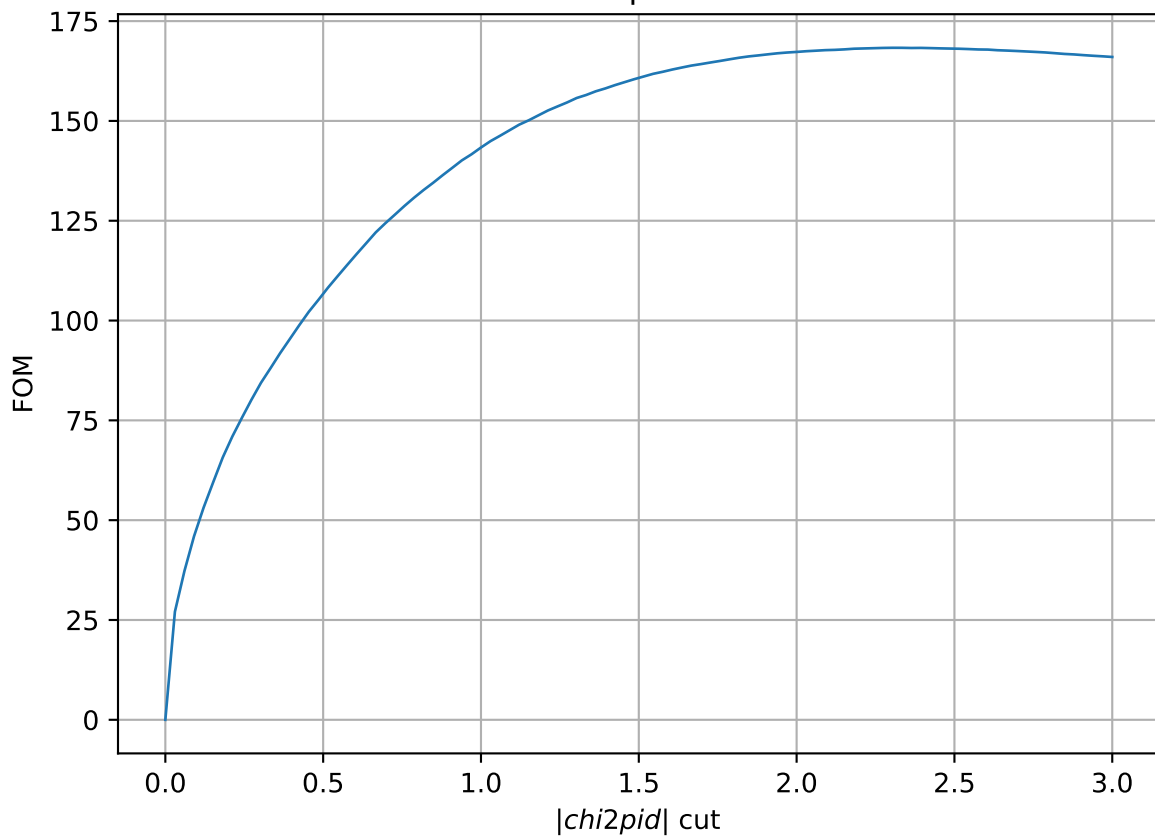
FOM vs $|\chi^2_{pid}|$ cut
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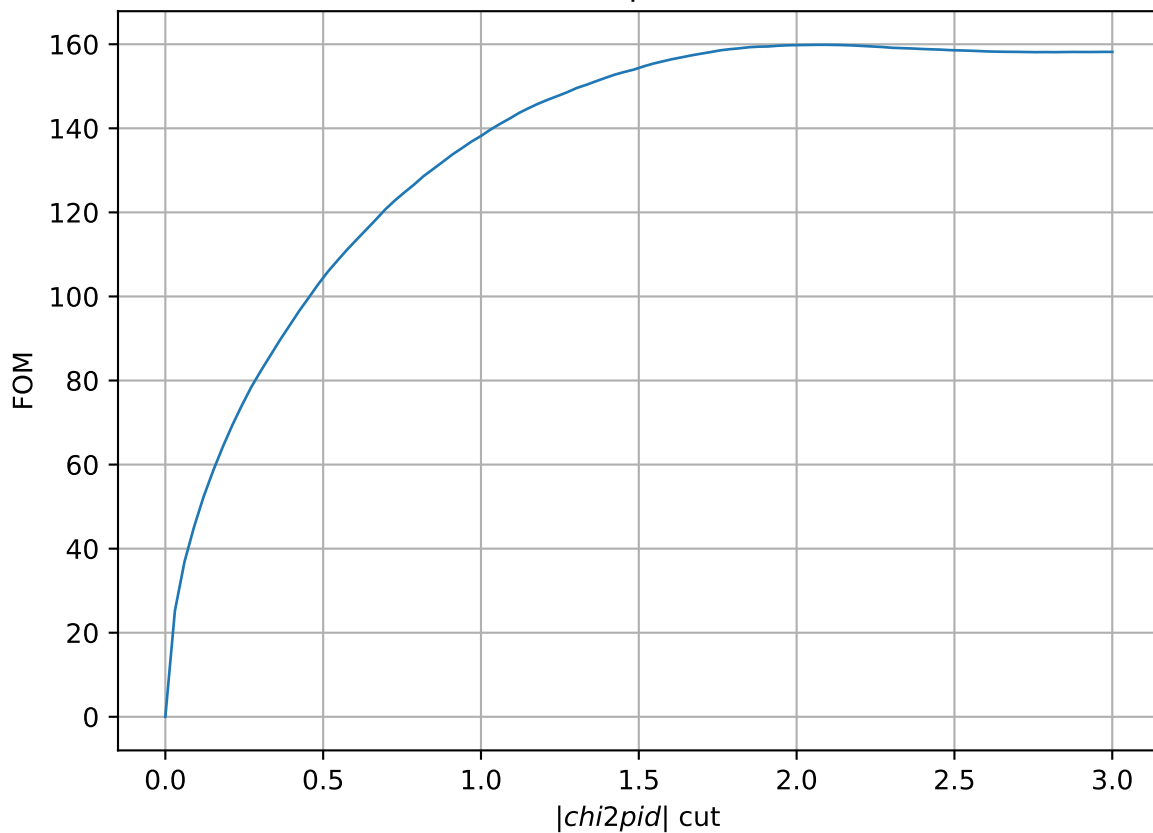
FOM vs $|\chi^2_{pid}|$ cut
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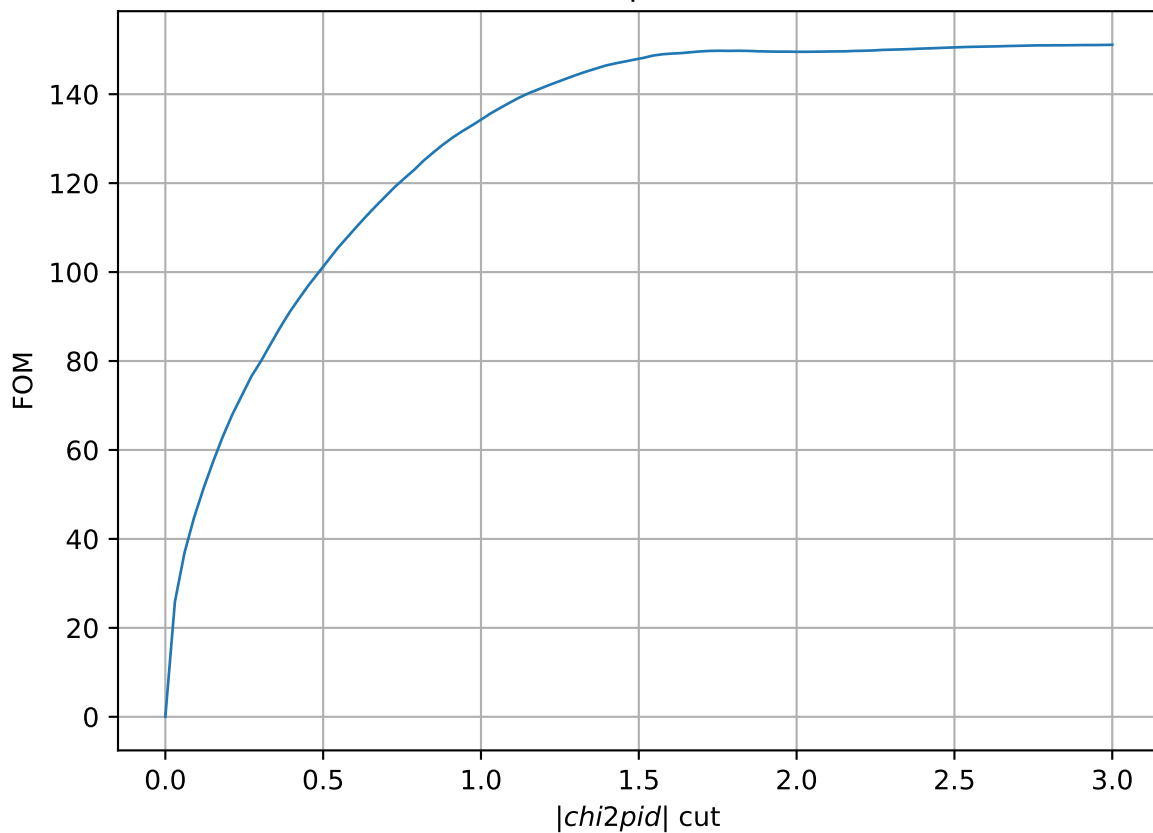
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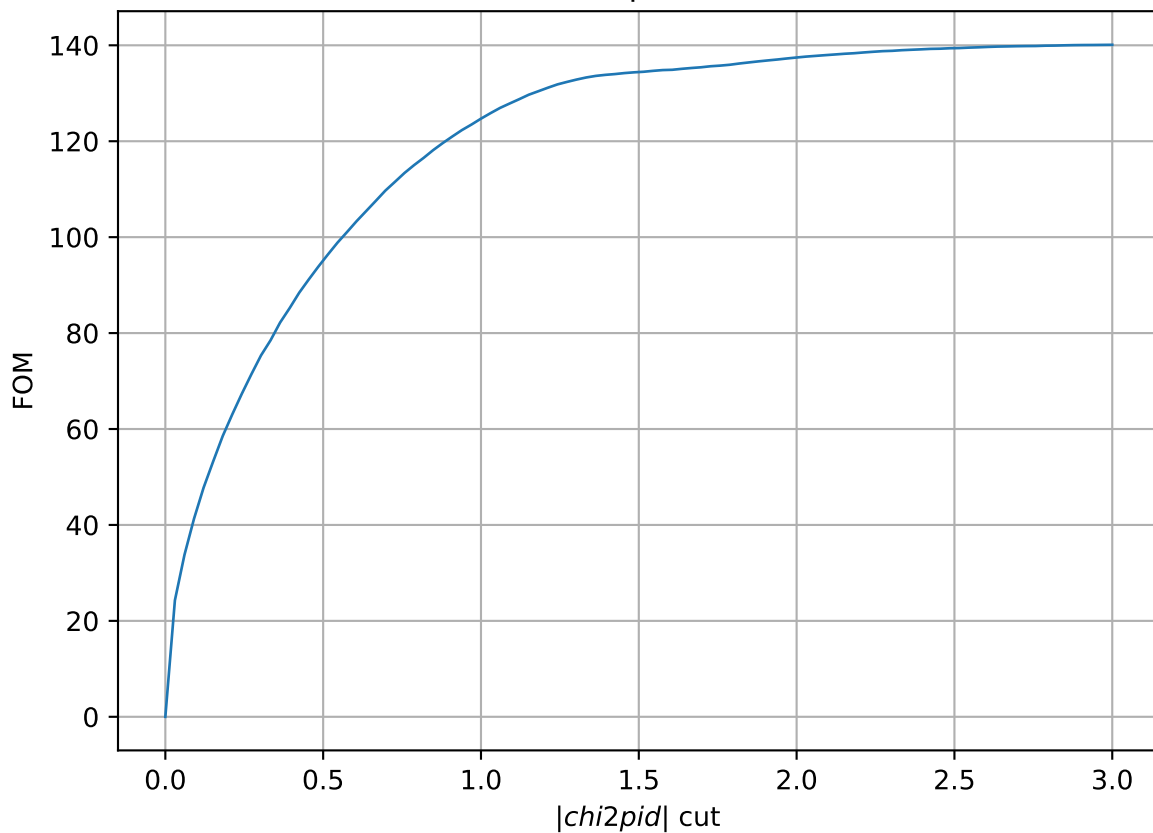
FOM vs $|\chi^2_{pid}|$ cut
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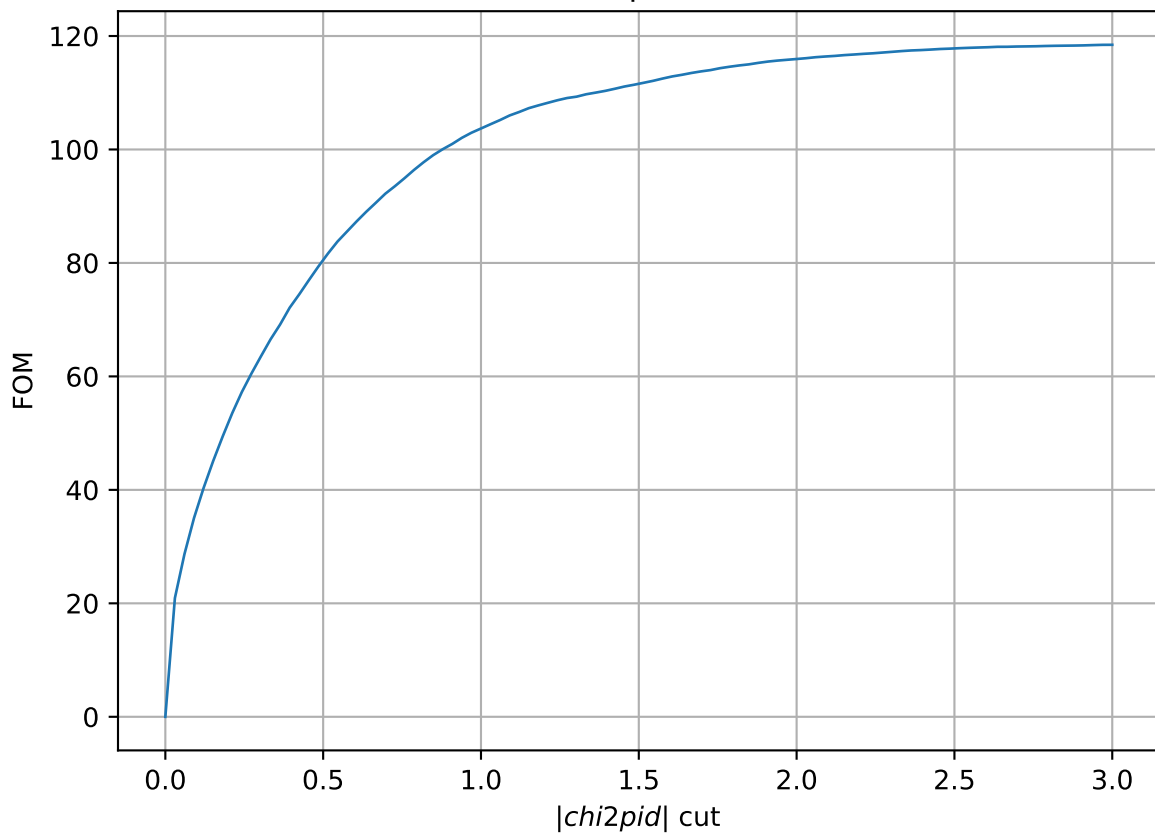
FOM vs $|\chi^2_{pid}|$ cut
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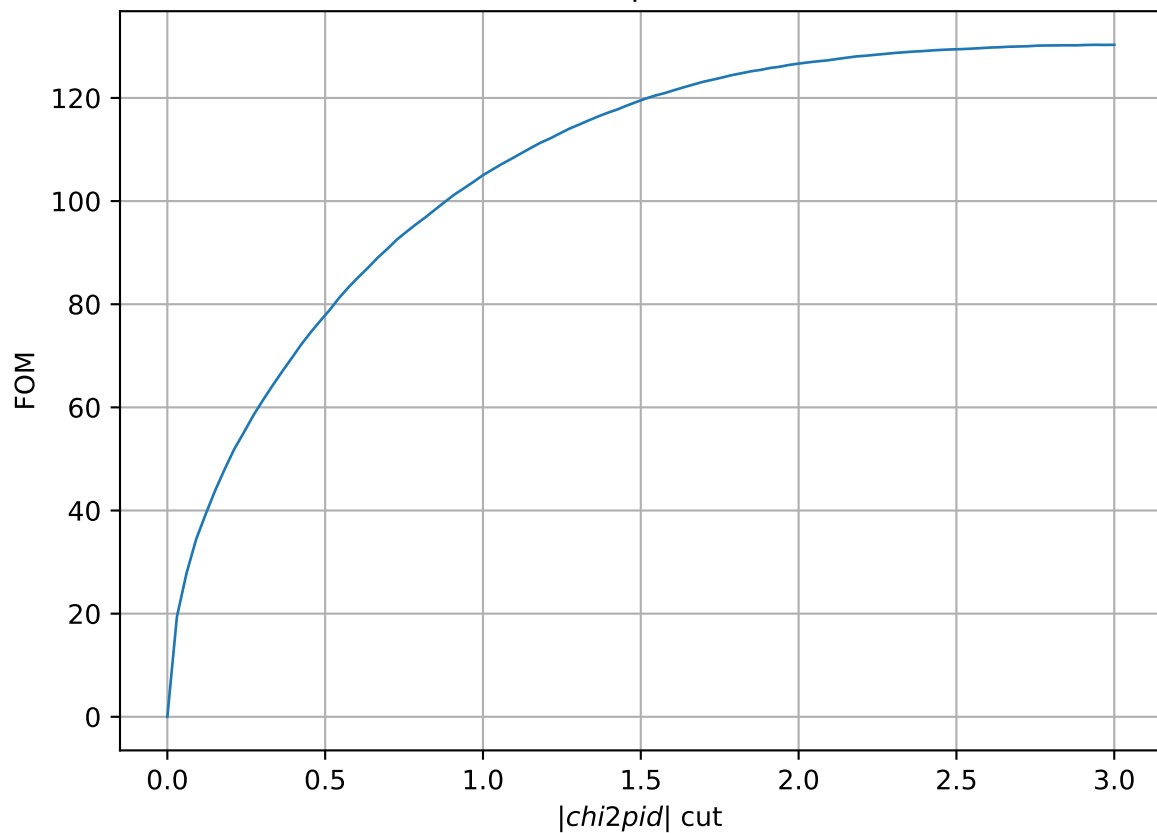
FOM vs $|\chi^2_{pid}|$ cut
 $23.0 < \theta < 29.0$
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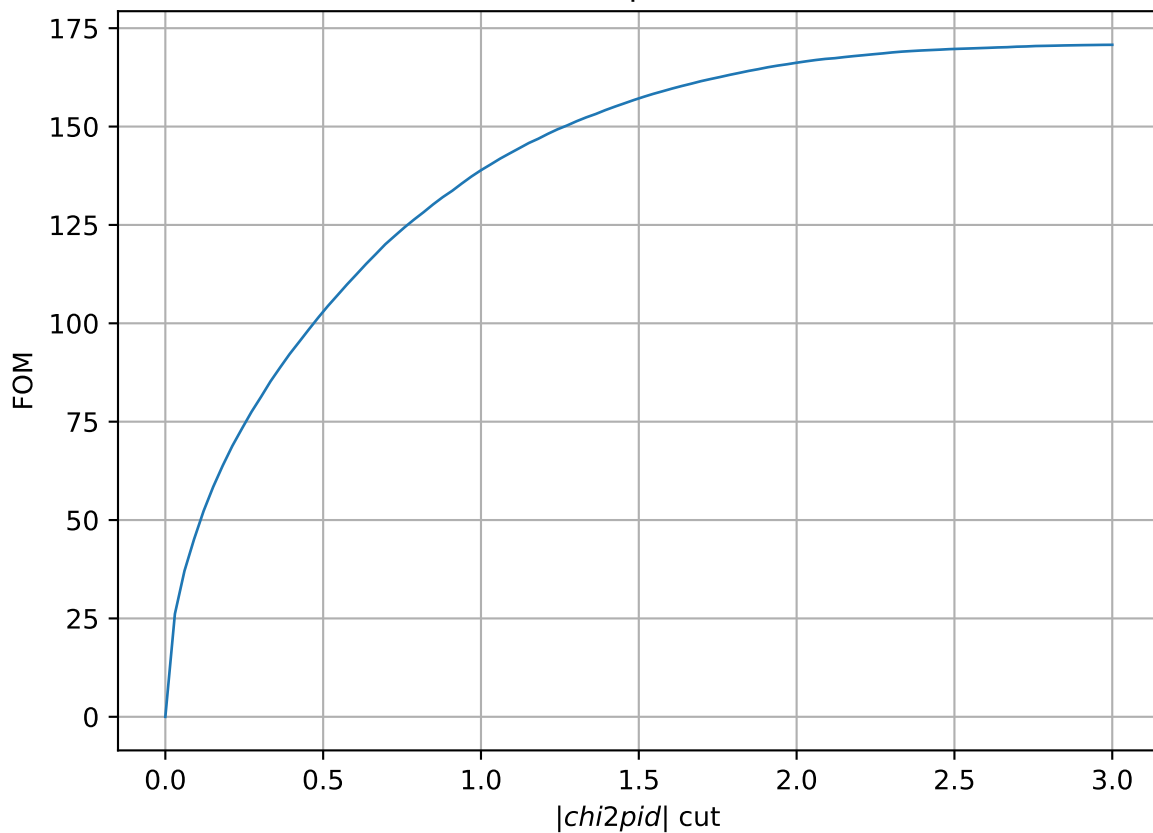
FOM vs $|\chi^2_{pid}|$ cut
 $23.0 < \theta < 29.0$
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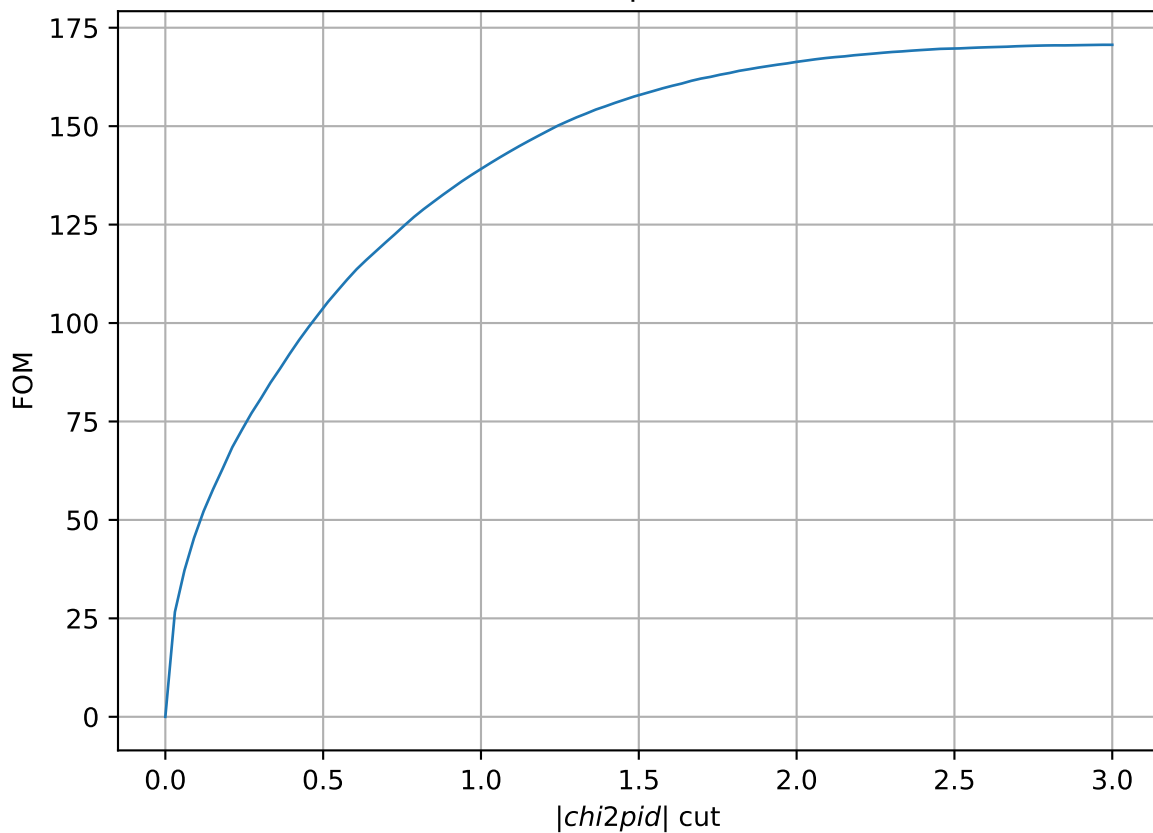
FOM vs $|\chi^2_{pid}|$ cut
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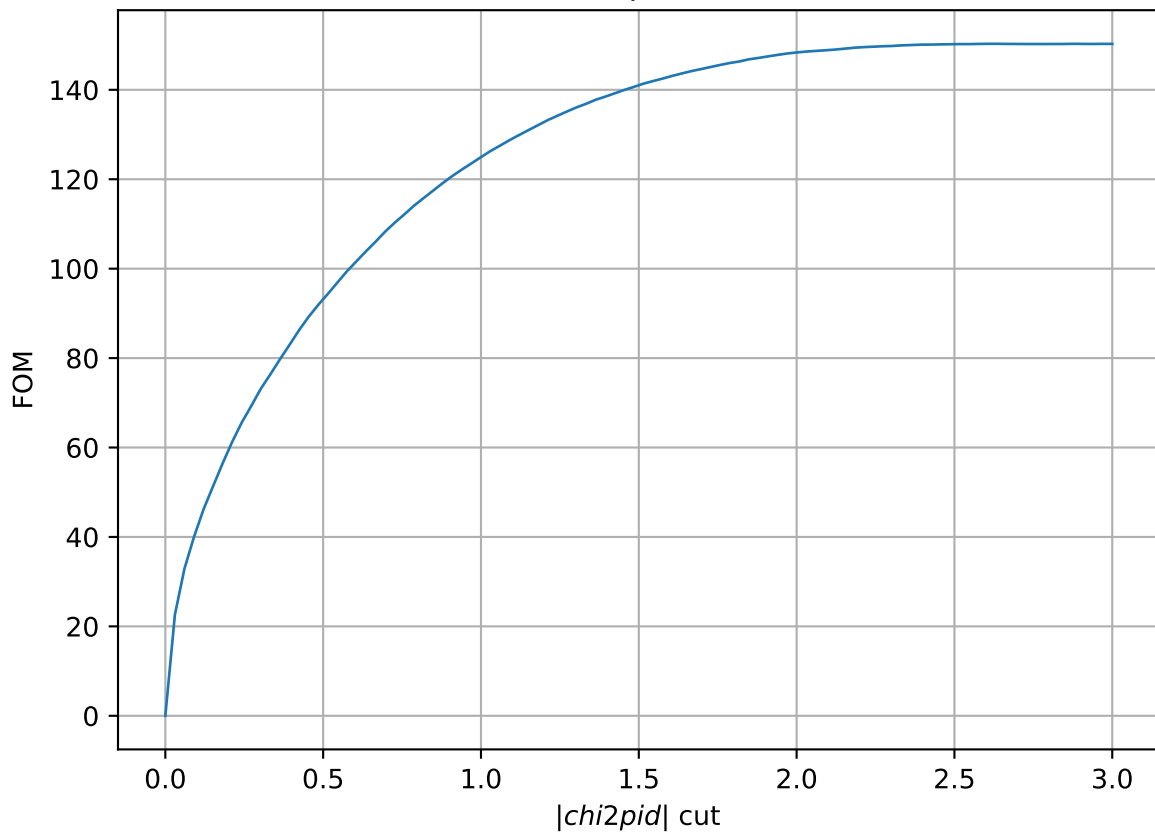
FOM vs $|\chi^2_{pid}|$ cut
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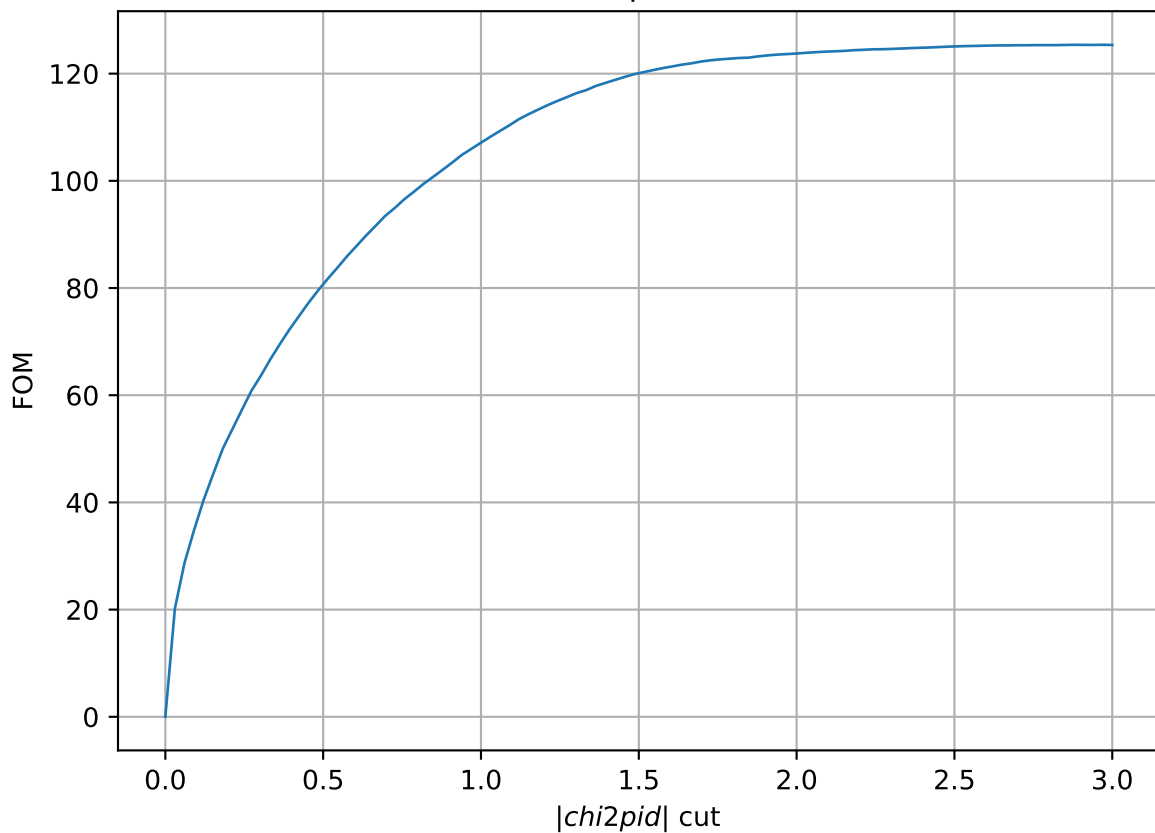
FOM vs $|\chi^2_{pid}|$ cut
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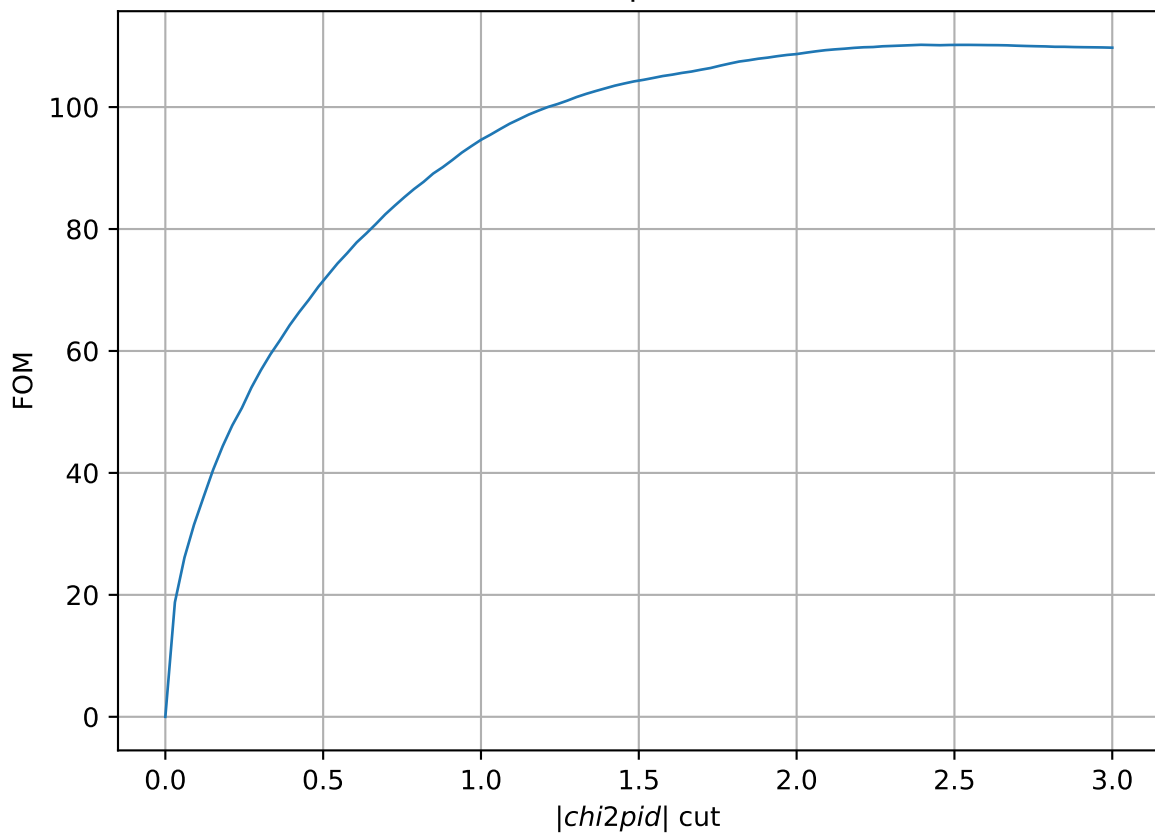
FOM vs $|\chi^2_{pid}|$ cut
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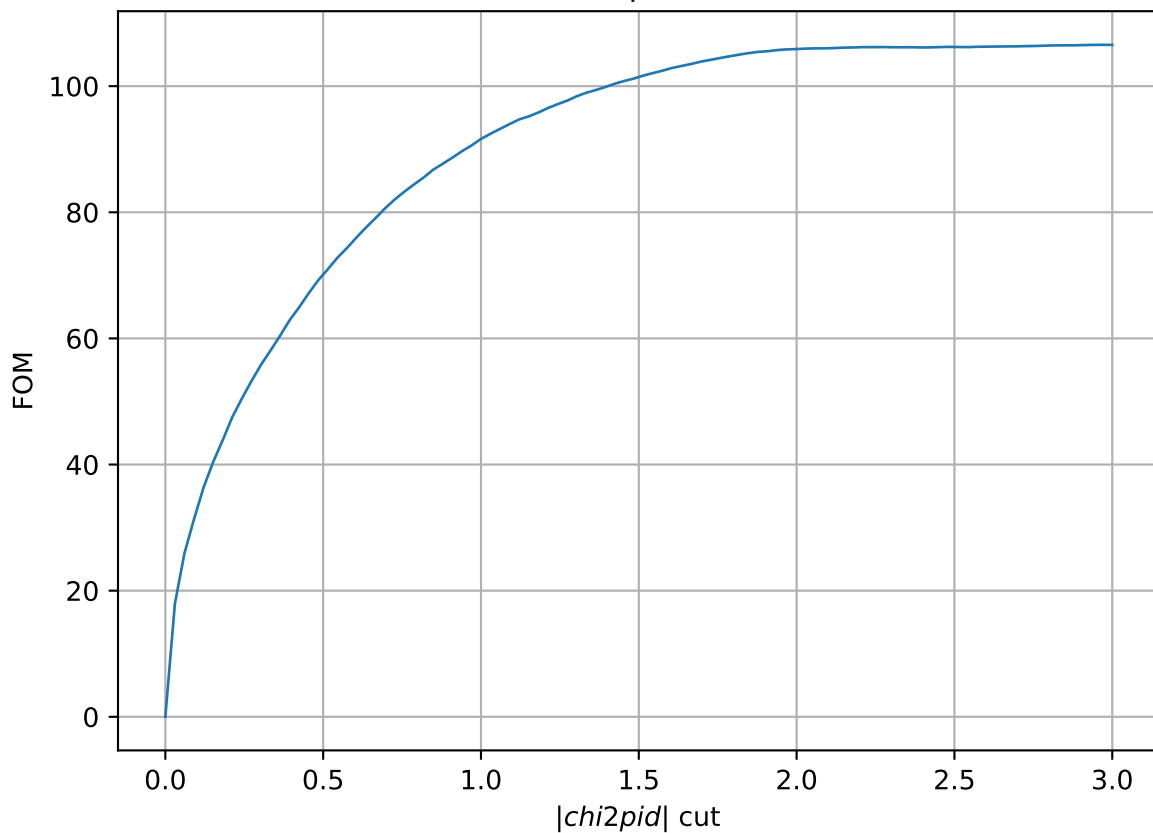
FOM vs $|\chi^2_{pid}|$ cut
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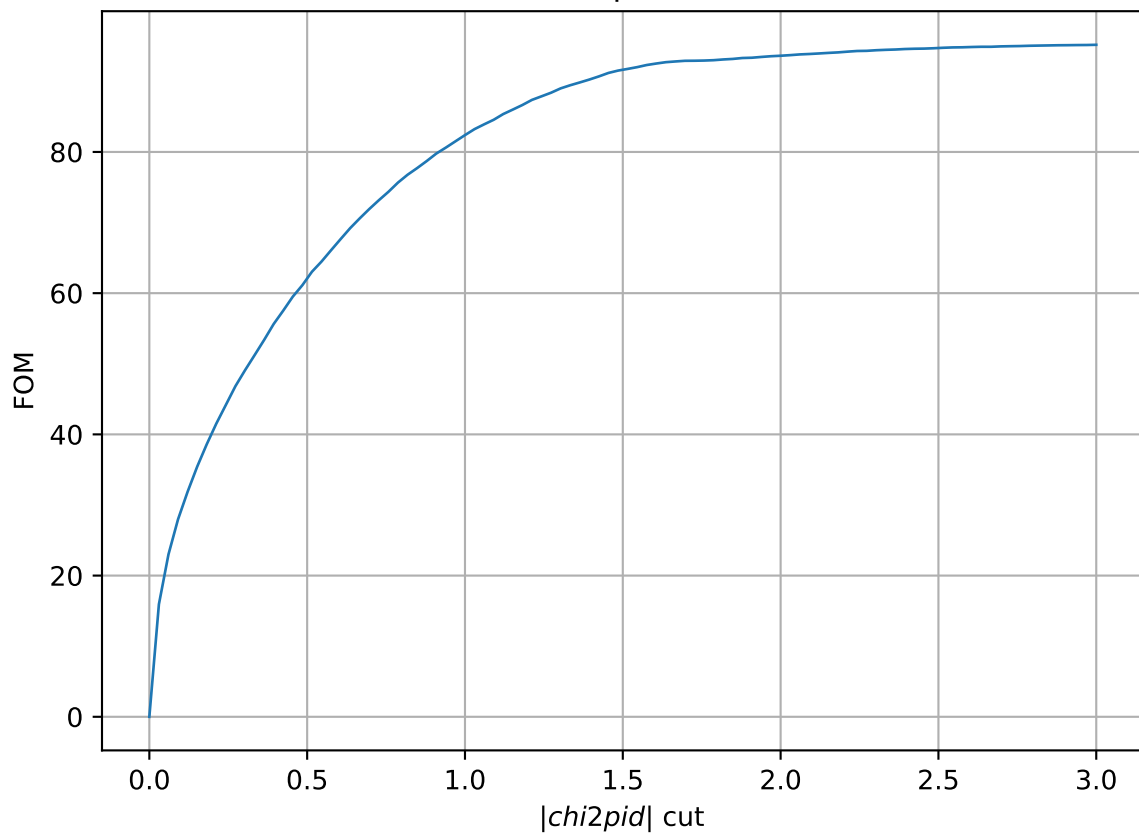
FOM vs $|\chi^2_{pid}|$ cut
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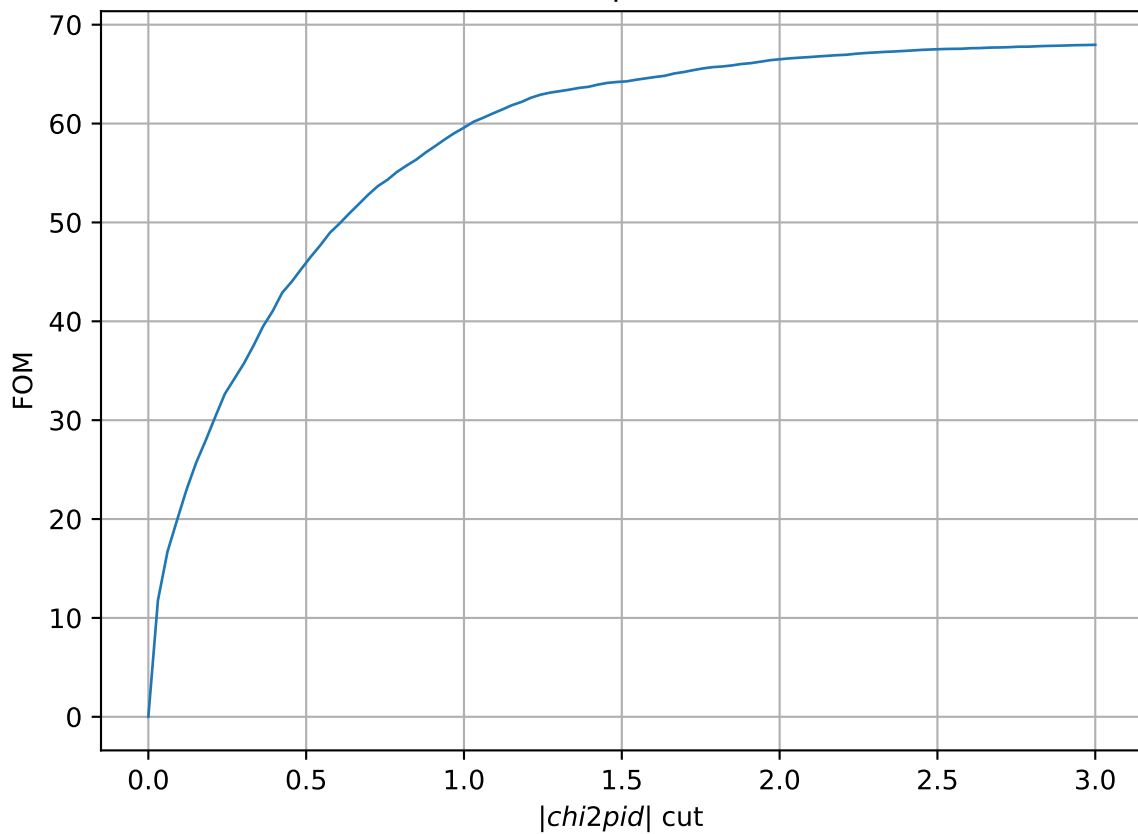
FOM vs $|\chi^2_{pid}|$ cut
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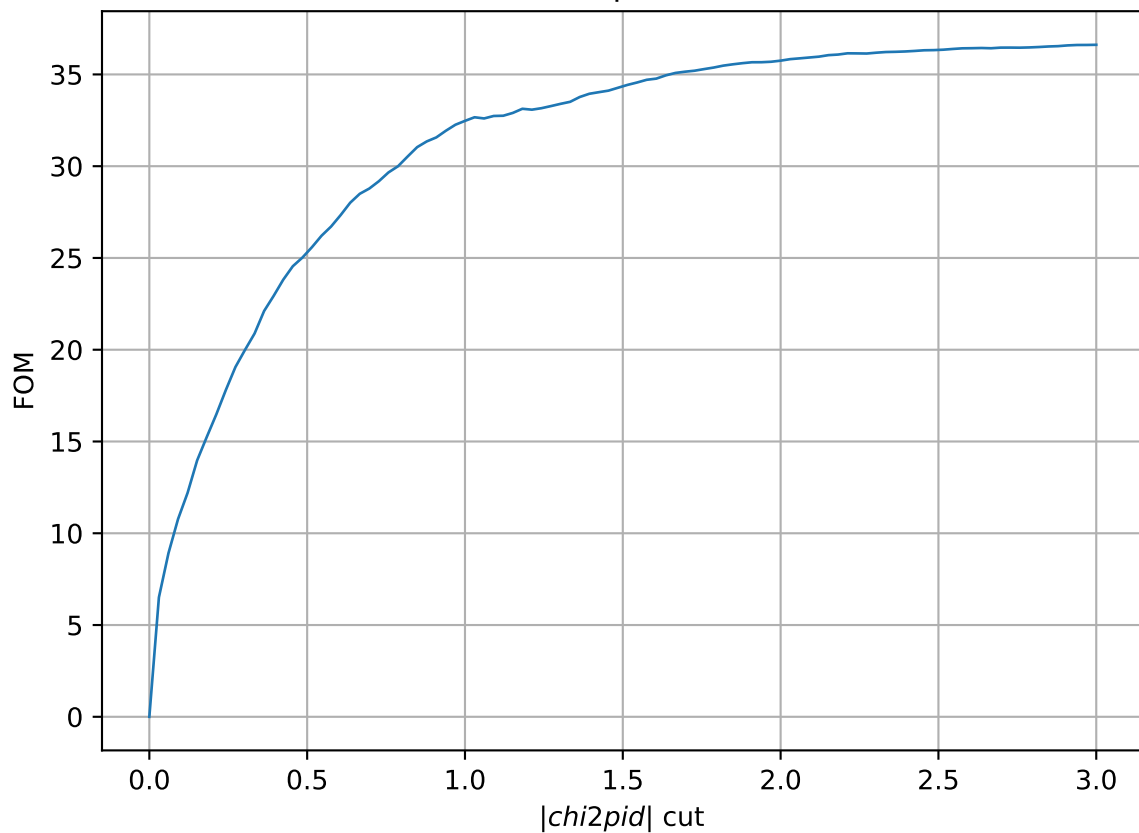
FOM vs $|\chi^2_{pid}|$ cut
 $29.0 < \theta < 35.0$
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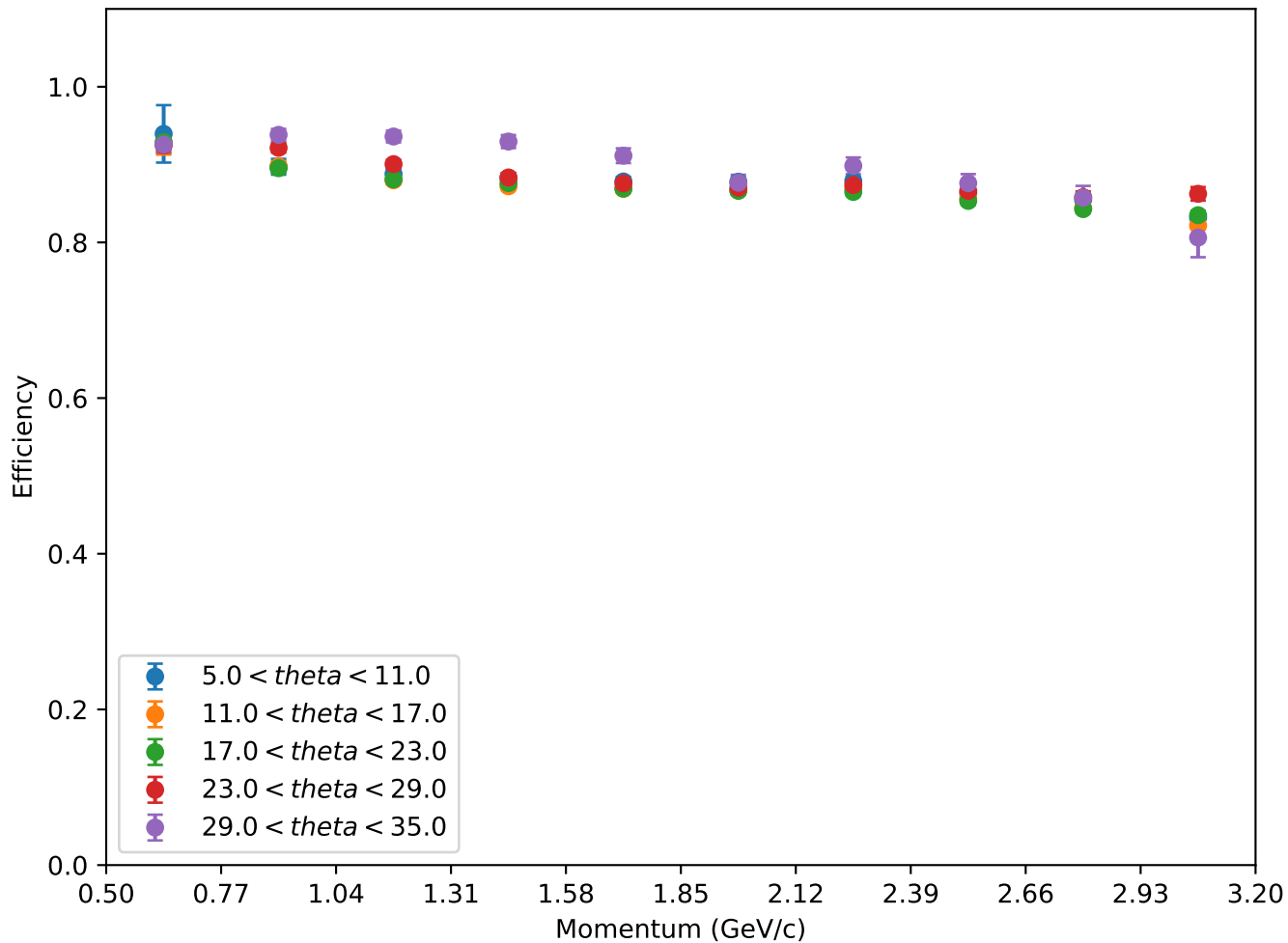
FOM vs $|\chi^2_{pid}|$ cut
 $29.0 < \theta < 35.0$
 $2.66 < p < 2.93$



FOM vs $|\chi^2_{pid}|$ cut
 $29.0 < \theta < 35.0$
 $2.93 < p < 3.20$



Efficiency (Using Optimized BDT)



Contamination (Using Optimized BDT)

